Optimal Design of Spot Instances in Cloud Services

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October 14, 2025

Abstract

We model spot cloud instances as posted contracts with a provider "kill option," summarized by an interruption hazard and a per-event refund. Buyers differ in how costly interruptions are; the provider trades current revenue against the option value of reclaiming capacity. We show the menu problem collapses to two sufficient statistics, the hazard and the cutoff type, so refunds serve only to implement, not create, value. We characterize optimal design, prove an equivalence to damaged-goods pricing, and map state-contingent hazards with refund schedules to simple priority auctions that deliver the same allocation and expected payments. A welfare decomposition reveals two distortions: relative to a welfare-maximizing optimum, too many buyers enroll in the preemptible class and, at that enrollment, reliability is set too high. We microfound salvage from congestion, derive comparative statics and policy tests for reliability caps and display rules, and propose a sufficient-statistics transfer based only on the hazard and the preemptible share that decentralizes the planner.

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1 Introduction

Major cloud service providers increasingly offer contracts that trade reliability for price. Under labels such as *preemptible*, *low-priority*, or *spot* instances, buyers accept the possibility of abrupt termination of service in exchange for substantial discounts relative to standard on-demand instances. While the setting is novel, the economic problem is classical: the seller uses a contract menu to screen demand heterogeneity and to marshal scarce capacity toward higher-value uses as conditions tighten. We argue that the cleanest abstraction for these offerings is a *spot* contract, a posted price agreement whose reliability is stochastically degraded by a provider-chosen interruption hazard and potentially softened by an explicit refund upon termination. Within this lens, the firm faces a dynamic trade-off between current revenue and the (state-contingent) salvage value of reclaimed capacity, and it can use the hazard and refund as levers to induce self-selection across heterogeneous customers' disutility from interruption.

Our framework is motivated by the institutional details of these markets. While early versions of spot instances featured real-time bidding, the market has since converged on a posted-price model. Providers now offer relatively smooth, deep discounts off a public on-demand price, explicitly stating that interruptions are not caused by competing bids but by the provider's own internal need to reclaim capacity[1]. Recent work by (Einav et al., 2018) using eBay data suggests that this trend is largely due to the fact that convenience and stable expectations come to dominate marginal gains from price discovery, with increasingly easier access to competitive information. These facts strongly motivate our choice to model the market as a posted-price menu, where the interruption hazard is a strategic policy variable tied to the provider's operational state, rather than a continuously cleared auction. On the demand side, the wide variety of workloads run on these instances, from fault-tolerant scientific computing to stateless web applications, provides clear evidence for the significant heterogeneity in interruption costs that lies at the heart of our screening model.

Our analysis connects three literatures that are often treated separately. First, we build on the theory of second-degree price discrimination and quality design. In monopoly screening models, quality is a design variable that sorts consumers by willingness to pay (Mussa and Rosen, 1978; Armstrong, 1996). For digital and information goods, "versioning" operationalizes this logic by creating quality/usage tiers that induce self-selection (Varian, 1997; Bhargava and Choudhary, 2008). We cast reliability as the quality attribute that is intentionally degraded for the lower-priced option, which places our environment in the damaged-goods tradition (Deneckere and McAfee, 1996). Recent work shows that low-cost quality adjustments and freemium line extensions continue to segment heterogeneous users (e.g., offering a lower-reliability tier), consistent with our interpretation (Lambrecht and Misra, 2017; Gu et al., 2018) . The novel feature here is that

[1] Amazon Web Service, 2018

degradation is not a fixed product choice but a *state-contingent* hazard under the seller's control, naturally aligning the sacrifice of low-tier reliability with moments in which capacity is most valuable elsewhere.

Second, the paper speaks to the queueing-and-priority pricing literature in operations and economics. Priority access and delay differentiation have long been used to align heterogeneous delay costs with service levels (Marchand, 1974; Mendelson and Whang, 1990; Ha, 2001). More recent work embeds these ideas in revenue-maximizing designs with guaranteed versus best-effort service, illuminating how prices and admission rules shape utilization and welfare (Allon and Federgruen, 2009; Maglaras et al., 2018). Directly related to our work, Dierks and Seuken (2022) combine queueing theory and game theory to demonstrate that there exists a simple condition under which a profit maximizing provider can offer both spot and fixed instances. Our mechanism, a posted price coupled with a state-dependent interruption hazard and a refund schedule, implements the same allocation logic as simple priority auctions, but via contracts that are native to cloud marketplaces and easier to scale. This equivalence provides both an interpretive bridge to classic priority pricing and a practical recipe for translating auction-based prescriptions into posted-price menus when market conventions or platform policy discourage bidding.

Third, we link to revenue management and dynamic pricing under capacity constraints. With perishable or reusable capacity and stochastic demand, optimal policies balance inventory/throughput against price responsiveness (Gallego and van Ryzin, 1994; Talluri and van Ryzin, 2004). In cloud markets, capacity is re-allocatable across instance families and regions, which creates a salvage value for interrupting the lowest-priority workloads. We show that the optimal hazard inherits the comparative statics familiar from revenue management: the hazard rises with the value of redeploying capacity and falls with the effective refund rate, with stronger effects when demand heterogeneity is wide enough to sustain separation. In this way, the kill-option framing is not only a screening device but also a dynamic control that reconciles value-based allocation with the realities of stochastic congestion.

Beyond theory, the institutional details of cloud spot markets motivate and discipline our modeling choices. Empirical studies of Amazon EC2's original spot market documented that spot prices were not allocatively revealing and often moved according to provider-side rules rather than continuous clearing of supply and demand (Agmon Ben-Yehuda et al., 2011). More recently, providers have emphasized posted discounts and price smoothing while explicitly reserving the right to interrupt, with documentation clarifying that spot prices adjust gradually with long-run capacity trends rather than instantaneous bids (Amazon Web Services, 2025). These facts suggest that posted menus with policy-controlled hazards are not merely a theoretical convenience; they are, in effect, the design space cloud platforms already inhabit. Our interruptible contract model formalizes the economic content of such policies and clarifies how refunds can substitute for blunt

reliability guarantees in restoring incentives and improving welfare.

Positioning within marketing and information systems underscores the broader relevance. Versioning and nonlinear pricing of digital goods highlight that low-cost quality adjustments can profitably segment demand (Sundararajan, 2004; Bhargava and Choudhary, 2008; Chellappa and Mehra, 2018). In service contexts, firms compete on both price and time guarantees, with capacity and delivery standards shaping market outcomes (Lederer and Li, 1997; So, 2000). Our environment blends these perspectives: reliability becomes a contracted service standard; the hazard is the instrument that ties the standard to congestion; and the refund is the inducement that preserves participation while maintaining screening power. Framed this way, preemptible instances become a textbook case of second-degree price discrimination with a queueing foundation, one that nests damaged-goods pricing as a limit (no state contingency) and priority auctions as an implementable benchmark (state-contingent rationing with incentive-compatible transfers).

Finally, we connect to adjacent domains where interruption rights are traded explicitly. Interruptible electricity tariffs and early-notification options similarly induce customers to self-select service priority in anticipation of shortfalls, using compensation to map private interruption costs into efficient rationing orders (Schweppe et al., 2013). Prior work has shown that interruptible contracts can mitigate supply problems associated with demand spikes from an electricity provider's of view (Baldick et al., 2006). Our results provide a microfoundation for analogous cloud contracts, with the advantage that the platform can set hazards in real time as capacity conditions evolve. Relative to auctions, posted spot menus can lower cognitive costs and stabilize expectations; relative to single-quality posted pricing, state-contingent degradation unlocks value when capacity is tight while maintaining a credible commitment to compensate displaced workloads. The remainder of the paper develops these ideas formally, establishing the cutoff selection rule, characterizing optimal hazards and refunds under canonical distributions, and detailing the equivalence to damaged-goods pricing and priority auctions under regularity.

Our contribution is threefold. First, we place the interruption hazard and the refund at the center of the contract, showing that a posted menu with a reclaim right is isomorphic to a vertical quality design in the spirit of information-goods versioning, yet disciplined by a capacity-redeployment motive that mirrors classic displacement costs in revenue management (Mussa and Rosen, 1978; Gallego and van Ryzin, 1994; Talluri and van Ryzin, 2004). This mapping brings the damaged-goods logic into a stochastic, operational setting: the provider optimally "damages" reliability when capacity is tight, and the compensating refund restores participation while preserving screening power (Deneckere and McAfee, 1996). Second, we establish an explicit equivalence between state-dependent interruption rules and simple priority mechanisms, connecting posted menus to the well-studied efficiency and implementability properties of pricing for priority in queues (Mendelson and Whang, 1990; Ha, 2001; Maglaras and Zeevi, 2005). This equivalence is actionable: it delivers testable

restrictions on how hazards and refunds should co-move with congestion and on how enrollment into the preemptible class should vary with observed discounts and service conditions.

Third, we provide a welfare decomposition that separates the enrollment margin from the hazard margin and yields clean sufficient statistics for regulation. The planner's target hinges on two primitives: the average interruption disutility among preemptible users and the marginal value of reallocating capacity. This perspective recovers well-known insights from priority pricing and service guarantees in operations and marketing, while offering a compact playbook for policy design in cloud settings where posted menus dominate auctions (Lederer and Li, 1997; So, 2000; Allon and Federgruen, 2009; Hassin and Haviv, 2003). Relative to advance-purchase and rationing instruments studied in peak-load and yield-management contexts, kill-option contracts tailor the rationing moment to realized congestion and make refunds the central redistributive lever (Dana, 1998; Gallego and van Ryzin, 1994; Talluri and van Ryzin, 2004).

Two empirical regularities motivate modeling choices and provide identification. Providers describe spot discounts as posted and relatively smooth, with interruptions governed by capacity headroom rather than high-frequency bidding. Historical evidence from the early EC2 spot market showed that instantaneous spot prices were not fully revealing of underlying supply-demand conditions, and later documentation codified a move toward posted discounts with explicit interruption rights (Agmon Ben-Yehuda et al., 2011; Amazon Web Services, 2025). These institutions favor posted menus over continuous auctions and make it natural to treat the hazard as a policy variable tied to congestion metrics. On the demand side, versioning results in extant literature suggest that reliable separation requires sufficient heterogeneity in interruption disutility; our model translates this requirement into a cutoff rule that can be taken to data using demand splits across contract classes.

Our framework also clarifies externalities and platform considerations. Priority pricing in queues highlights how charging for faster service internalizes delay costs; in cloud markets, interruption hazards may spill over to guaranteed users via shared control planes or to the broader ecosystem via performance risk. Classic results on efficiency loss in network resource allocation caution that price-only mechanisms can leave residual wedges when congestion externalities are not fully priced (Johari and Tsitsiklis, 2004). We therefore complement the private-optimum analysis with a welfare benchmark and simple instruments that operate on observables like the hazard and the share of buyers self-selecting into the preemptible tier. The intent is pragmatic: to provide tools that are robust to whether a provider uses auctions, posted menus, or hybrids, and that can be implemented using telemetry and A/B experiments rather than full structural models.

The paper proceeds by laying out the model and its primitives, establishing the cutoff rule and the optimal design of hazards and refunds, and then mapping the preemptible contract to both damaged-goods pricing and priority auctions. We next develop the welfare decomposition and

compare monopoly and planner allocations, before analyzing policy instruments such as reliability caps, display constraints on posted discounts, and sufficient-statistics transfers keyed to the hazard and enrollment share. Throughout, we aim for results that are diagnostically useful: comparative statics that align with how operators value headroom, identification strategies that rely on changes in posted discounts and observed interruption rates, and welfare criteria that can be expressed without heavy notation. In this way, the model connects established ideas in screening, priority pricing, and revenue management to the distinctive institutional details of cloud markets, where contracts must scale, interfaces must be simple, and risk must be allocated.

2 A Model of Kill-Option Contracts

We develop a model of a monopolistic cloud provider designing a menu of service contracts to screen a population of heterogeneous buyers. The core of the design problem lies in trading off price against reliability, where reliability is not a fixed attribute but a state-contingent operational choice controlled by the provider. The framework is designed to capture the essential economics of preemptible or "spot" instances, where customers accept a lower price in exchange for the risk of service interruption, and the provider gains a valuable option to reclaim capacity when it is most needed.

2.1 Primitives: Buyers, Contracts, and Salvage Value

Buyers and Preferences. We consider a market with a continuum of potential buyers, normalized to a unit mass. Each buyer seeks to procure one unit of a computational resource (an "instance") for a normalized period of time. All buyers derive a common gross utility, $v > 0$, from the successful, uninterrupted completion of their task. This common value represents the baseline productivity of the cloud service, which we assume is homogeneous across users to focus squarely on heterogeneity in interruption costs.

Buyers are distinguished by a single dimension of private information: their sensitivity to service interruption, which we denote by φ. This parameter, which we refer to as a buyer's "type," represents the full economic disutility incurred if the service is terminated prematurely. This cost can encompass a wide range of factors, such as the direct engineering effort required to checkpoint and restart a job, the value of lost computational state, the opportunity cost of project delays, or the reputational harm from failing to meet a service-level objective. We assume φ is distributed across the buyer population according to a continuously differentiable cumulative distribution function (CDF) $F(\cdot)$ with a strictly positive probability density function (PDF) $f(\cdot)$ on its support $[\underline{\varphi}, \bar{\varphi}]$, where $0 \leq \underline{\varphi} < \bar{\varphi}$. The assumption of a continuous distribution is standard and ensures that no single buyer type has a discrete impact on market outcomes. Buyers are assumed to be risk-neutral and seek to maximize their expected utility. The outside option of not purchasing any service

yields zero utility. This heterogeneity is a defining feature of the market; workloads commonly described as suitable for spot instances, such as batch data processing, one-off data analytics, genomic sequencing, or CGI rendering, are by design fault-tolerant and thus correspond to a low φ. In contrast, critical workloads like relational databases or real-time transaction processing systems are highly sensitive to interruption and represent a high φ.

Contracts and the Provider's Choices. The provider (seller) posts a menu consisting of two contracts, a standard approach for screening a market with two unobserved segments (in our case, those with "high" and "low" interruption costs). The menu is designed to induce buyers to self-select based on their private information φ.

- **Guaranteed Contract (G)**: This is a premium, fully reliable service. It is defined by a single parameter: a price $p_G \geq 0$. By choosing this contract, the buyer receives an assurance of zero interruptions during the service window.
- **Preemptible Contract (P)**: This is a discounted, lower-reliability service, which we model as a "kill-option" contract. It is defined by a triplet of choice variables for the provider: (p_P, λ, r), where $p_P \geq 0$ is the price, $\lambda \in [0, 1]$ is the interruption hazard, and $r \geq 0$ is a refund paid to the buyer in the event of an interruption.

The interruption hazard, λ, is a key policy instrument for the provider. We interpret it as the ex-ante probability of at least one service termination event during the usage window. This can be microfounded as the integrated hazard from an underlying stochastic process; for instance, if interruptions arrive according to a Poisson process with an instantaneous rate $h(t)$, then $\lambda \approx \int_0^1 h(t)dt$ for small rates. More generally, we can think of λ as the expected number of compensated interruption events over the period. This formulation ensures that the expected disutility from interruptions is linear in both the buyer's type φ and the provider's choice of hazard λ. The refund r is specified on a per-event basis, so the total expected compensation for interruptions is $r\lambda$.

Buyer and Seller Payoffs. Given these definitions, the expected utility for a buyer of type φ from choosing each contract is straightforward. The guaranteed contract offers a deterministic payoff, while the preemptible contract involves an expectation over the random interruption event:

$$U_G(\varphi) = v - p_G, \qquad U_P(\varphi) = v - p_P - \varphi\,\lambda + r\,\lambda. \tag{1}$$

The term $\varphi\lambda$ captures the expected cost of interruption for type φ, while $r\lambda$ is the expected compensation.

On the seller's side, we normalize the marginal operating costs of providing an instance to zero for both contract types. This is a common and reasonable assumption for digital services where

the primary costs are fixed capacity investments, and the marginal cost of serving one more user is negligible. The key operational feature we aim to capture is that interrupting a preemptible instance frees up capacity that can be repurposed for other valuable uses. This could include serving unexpected bursts of on-demand requests, handling high-priority internal workloads, or avoiding the need to provision additional expensive capacity during peak times. We summarize the expected value generated by reclaiming this capacity in a *salvage function*, $b(\lambda)$. This function maps the chosen hazard level into an expected revenue or cost-saving for the provider. We impose the following standard economic assumptions on its shape:

$$b : [0, 1] \to \mathbb{R}_+, \quad b(0) = 0, \quad b'(\lambda) \geq 0, \quad \text{and} \quad b''(\lambda) \leq 0. \tag{2}$$

The condition $b(0) = 0$ is a normalization: a zero hazard yields no interruptions and hence no salvage. The assumption that $b(\cdot)$ is increasing ($b' \geq 0$) reflects that a higher interruption hazard frees up more capacity in expectation, generating more value. The crucial assumption is concavity ($b'' \leq 0$), which captures diminishing marginal returns to reclaiming capacity. As we will microfound in Section 2.4, this arises naturally if the provider strategically targets interruptions to occur during the most congested periods; the first interruptions free up capacity when it is most valuable, while subsequent interruptions occur in progressively less critical states.

The seller's expected profit per buyer for each contract is therefore:

$$\pi_G = p_G, \qquad \pi_P = p_P + b(\lambda) - r\,\lambda. \tag{3}$$

The provider's problem is to choose the menu of contracts (p_G, p_P, λ, r) to maximize total expected profit, subject to buyers' self-selection and participation decisions. The timing is as follows: (1) The seller posts the menu. (2) Buyers observe the menu and choose a contract to maximize their expected utility. (3) Service commences, congestion states realize, and interruptions occur according to the hazard λ. (4) Payoffs are realized. Table 1 summarizes the key notation.

2.2 Buyer Self-Selection and the Cutoff Rule

The effectiveness of the menu hinges on how buyers sort themselves between the two contracts. A buyer of type φ prefers the guaranteed option if $U_G(\varphi) \geq U_P(\varphi)$. The utility difference, $\Delta(\varphi) = U_G(\varphi) - U_P(\varphi)$, governs this choice:

$$\Delta(\varphi) = (v - p_G) - \big(v - p_P - \varphi\lambda + r\lambda\big) = (p_P - p_G) + (\varphi - r)\lambda.$$

For any positive hazard $\lambda > 0$, the term $\Delta(\varphi)$ is strictly increasing in the buyer's type φ. This linear structure gives rise to a clean single-crossing property: the net utility from choosing the

Table 1: Notation for Primitives and Contracts

Symbol	Meaning
$v > 0$	Common gross value for uninterrupted service
$\varphi \in [\underline{\varphi}, \bar{\varphi}]$	Interruption sensitivity (disutility); buyer's type
F, f	CDF and density of φ
$G = (p_G, 0)$	Guaranteed contract: price p_G, zero hazard
$P = (p_P, \lambda, r)$	Preemptible contract: price p_P, hazard λ, refund r
U_G, U_P	Buyer utilities under each contract
$b(\lambda)$	Expected salvage value from reclaiming capacity with hazard λ
π_G, π_P	Per-buyer profits under each contract

more reliable (guaranteed) option increases with the buyer's sensitivity to interruption. This is the standard sorting condition in screening models, and it ensures that the market will be partitioned by a simple threshold rule.

Specifically, there will exist a unique cutoff type, $\varphi^\star$, who is exactly indifferent between the two contracts, i.e., for whom $\Delta(\varphi^\star) = 0$. Solving for this cutoff yields its closed-form expression in terms of the menu parameters:

$$\varphi^\star \;=\; \frac{p_G - p_P}{\lambda} + r. \tag{4}$$

All buyers with interruption sensitivity lower than this threshold ($\varphi < \varphi^\star$) find the price discount of the preemptible option sufficient to compensate for the risk of interruption and will choose contract P. Conversely, all buyers with higher sensitivity ($\varphi \geq \varphi^\star$) will opt for the safety of the guaranteed contract G. Because the type distribution F is continuous, the set of indifferent buyers at $\varphi^\star$ has measure zero, so our tie-breaking rule (that indifferent buyers choose G) is inconsequential for aggregate outcomes. The market shares are therefore $D_P = F(\varphi^\star)$ for the preemptible class and $D_G = 1 - F(\varphi^\star)$ for the guaranteed class.

The cutoff formula (4) reveals the mechanics of market segmentation. The provider can increase the size of the preemptible market (raise $\varphi^\star$) by either widening the price gap $p_G - p_P$ or by offering a more generous refund r. Conversely, for a fixed price-refund structure, increasing the hazard λ makes the preemptible option less attractive, pushing more sensitive buyers toward the guaranteed product and thus lowering $\varphi^\star$. This highlights the fundamental substitutability between the price discount and the refund in managing demand.

2.3 The Seller's Problem: Reduction to Sufficient Statistics

The seller's objective is to choose the four menu parameters (p_G, p_P, λ, r) to maximize total expected profit. This problem appears complex due to the high dimensionality of the choice space. However, the structure of buyer self-selection allows for a powerful simplification. The key insight

is that the seller does not care about the specific values of p_P and r independently, but only about the cutoff $\varphi^\star$ they jointly induce. We can use the indifference condition (4) to eliminate one of these variables. Solving for p_P, we have $p_P = p_G - \lambda(\varphi^\star - r)$. Substituting this into the profit expression, $\Pi = \big(1 - F(\varphi^\star)\big)p_G + F(\varphi^\star)\big(p_P + b(\lambda) - r\lambda\big)$, yields:

$$\begin{aligned}\Pi &= \big(1 - F(\varphi^\star)\big)p_G + F(\varphi^\star)\Big(\big(p_G - \lambda\varphi^\star + r\lambda\big) + b(\lambda) - r\lambda\Big) \\ &= p_G + F(\varphi^\star)\Big(b(\lambda) - \varphi^\star\lambda\Big).\end{aligned}$$

This reduced form of the profit function is parsimonious and is central to our analysis. It reveals two critical facts. First, the refund r has completely dropped out of the expression. This confirms that its role is purely *implementational*: for any target hazard λ and cutoff $\varphi^\star$, the seller can use a continuum of (p_P, r) pairs that satisfy the indifference condition[2]. These pairs will all yield the same market segmentation, the same buyer utilities, and the same total profit. The refund is not a fundamental source of value but a flexible instrument for achieving the desired price gap.

Second, the guaranteed price p_G enters the profit function additively. To maximize profit, the seller should set p_G as high as possible, subject to the participation constraint of the buyers who select the guaranteed contract. Since their utility is $U_G = v - p_G$, participation requires $p_G \leq v$. The seller will therefore optimally set $p_G^\star = v$, extracting the full surplus from all buyers in the guaranteed segment. These two steps reduce the seller's complex four-dimensional problem to a much simpler two-dimensional one.

Proposition 2.1 (Program Reduction and Implementability)**.** *The seller's problem of choosing an optimal menu is equivalent to choosing a hazard $\lambda \in [0, 1]$ and a cutoff type $\varphi^\star \in [\underline{\varphi}, \bar{\varphi}]$ to solve:*

$$\max_{\lambda, \varphi^\star} \ \Pi(\varphi^\star, \lambda) = v + F(\varphi^\star)\big(b(\lambda) - \varphi^\star\lambda\big). \tag{5}$$

Any solution $(\lambda^\star, \varphi^\star)$ can be implemented by setting $p_G^\star = v$ and choosing any pair (p_P, r) that satisfies $p_P = v - (\varphi^\star - r)\lambda^\star$.

The objective function in (5) has a clear economic interpretation. The seller's profit above the baseline of selling a guaranteed product to everyone for price v is determined by the term $F(\varphi^\star)\big(b(\lambda) - \varphi^\star\lambda\big)$. This is the product of the size of the preemptible market, $F(\varphi^\star)$, and the net surplus extracted from each buyer in that market. This surplus consists of the salvage value $b(\lambda)$

[2]This theoretical property that refunds are secondary to the upfront discount is consistent with market practice. Providers typically do not offer explicit cash refunds. Instead, the "refund" is an implicit forgiveness of charges for the partial billing period in which the interruption occurred. For example, a user might not be charged for the final, incomplete hour of a terminated instance. This practice corresponds to a minimal, administratively simple refund, with the primary incentive for adoption being the large posted discount, just as our model would suggest.

minus the information rent, $\varphi^\star \lambda$, that must be left to the marginal type $\varphi^\star$ to ensure her participation in the preemptible class.

2.4 A Microfoundation for Salvage Value

The salvage function $b(\lambda)$ plays a critical role in this framework, as its shape, particularly its concavity, determines the optimal reliability policy. We now provide a concrete microfoundation for this function based on the provider's operational problem of managing capacity under fluctuating congestion.

Let $s \in \mathcal{S}$ be the state of the system, capturing all relevant factors like demand from other services, energy prices, or network load. Let $\mu(s) \geq 0$ denote the shadow value of one unit of capacity in state s. This is the marginal value to the provider of having an additional instance available, which can be thought of as the Lagrange multiplier on the capacity constraint in the provider's global optimization problem. A high $\mu(s)$ signifies a state of high congestion where capacity is scarce and valuable.

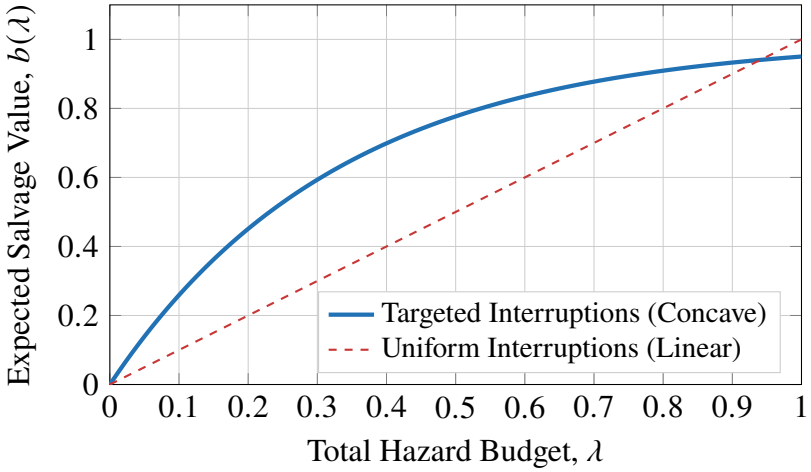


Figure 1: Illustrative salvage functions. With targeted interruptions, the provider "skims" the highest-value states first, leading to a strictly concave frontier. With uniform (non-targeted) interruptions, the value is linear with a slope equal to the average shadow value of capacity.

The provider's interruption policy is a state-contingent hazard schedule, $h(s) \in [0, 1]$. For a given total hazard budget λ, the provider's goal is to design the schedule $h(s)$ to maximize the expected salvage value. This is a targeting problem:

$$b(\lambda) \equiv \sup_{h(\cdot):\, 0 \leq h(s) \leq 1,\, \int h(s) m(ds) = \lambda} \int_{\mathcal{S}} \mu(s)\, h(s)\, m(ds). \tag{6}$$

The economic logic for solving this problem is clear: to get the most value from a limited budget of interruptions, the provider should prioritize them in states where capacity is most valuable. This

leads to a simple threshold policy where interruptions are triggered if and only if $\mu(s)$ exceeds some cutoff $\kappa(\lambda)$. This "cream-skimming" of high-μ states leads directly to diminishing marginal returns, hence $b''(\lambda) \leq 0$, as illustrated in Figure 1. If the provider were unable to target interruptions, the salvage function would be linear. The ability to dynamically target interruptions is thus the operational source of the concavity that shapes the optimal menu design.

Figure 2 provides a simple illustration of this principle. If the shadow value of capacity can take on only two values, high (μ_H) or low (μ_L), the optimal interruption policy is to first exhaust all high-value states before beginning to interrupt in low-value states. The resulting salvage function $b(\lambda)$ is piecewise linear and concave, with a kink at the point where the high-value states are exhausted. The marginal salvage, $b'(\lambda)$, is a step function, vividly demonstrating the principle of diminishing returns to reclamation. This microfoundation provides a direct economic interpretation for the marginal salvage value $b'(\lambda)$: it is the shadow value of capacity in the marginal state where an interruption would occur. This interpretation will be crucial when we characterize the optimal menu in the next section.

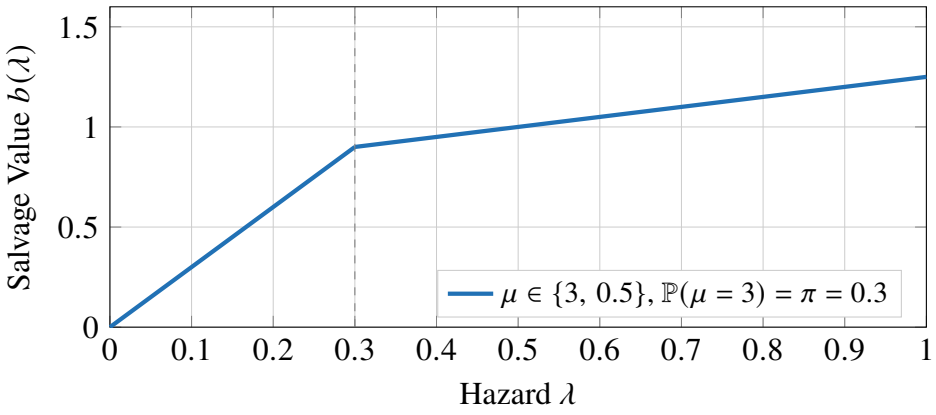
3 Optimal Design and Comparative Statics

Having established in Section 2 that the provider's complex menu design problem collapses to a choice over two sufficient statistics, the interruption hazard λ and the market-segmenting cutoff $\varphi^\star$, we now turn to the formal characterization of the optimal menu. This section dissects the provider's optimization problem to uncover the core economic principles governing the trade-off between reliability, market segmentation, and the operational value of reclaimed capacity. Our analysis will proceed in stages. We first characterize the economically rich interior solution, where the provider finds it optimal to offer a genuinely risky preemptible contract that appeals to a non-trivial segment of the market. We then provide a comprehensive analysis of the boundary solutions that arise when the underlying economic trade-offs are more extreme, ensuring our characterization is complete. This detailed analysis will form the foundation for deriving comparative statics and, in subsequent sections, for understanding the welfare properties of the market outcome.

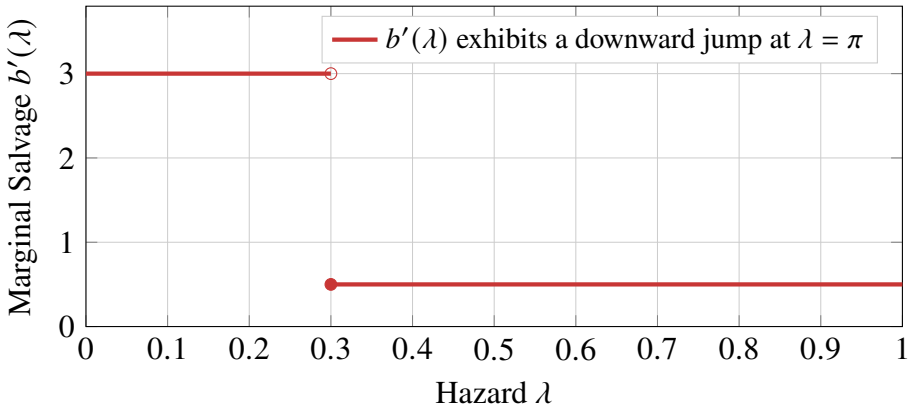
3.1 Interior Characterization (Regular Types)

We begin our inquiry by focusing on the conditions that define an interior optimum, where the provider's chosen allocation $(\varphi^\star, \lambda)$ lies strictly within the feasible set, i.e., $\lambda \in (0,1)$ and $\varphi^\star \in (\underline{\varphi}, \bar{\varphi})$. This scenario corresponds to a market structure where both the guaranteed and preemptible contracts coexist and each attracts a positive mass of buyers. The provider's objective, as derived in Proposition 2.1, is to solve the following reduced program:

$$\max_{\lambda\in[0,1],\ \varphi^\star\in[\underline{\varphi},\bar{\varphi}]} \Phi(\varphi^\star, \lambda) = F(\varphi^\star)\big(b(\lambda) - \varphi^\star\lambda\big),$$



(a) Salvage value $b(\lambda)$ is piecewise linear with a kink at $\lambda = \pi$.



(b) Marginal salvage $b'(\lambda)$ is a step function: high in the top states, low afterward.

Figure 2: Salvage Function from a Two-Point Mixture of Shadow Values. The operator interrupts in high-value states (mass π) first, then in low-value states.

where we recall that F is the CDF of buyer types and b is the salvage value function, which we assume to be increasing and weakly concave. For an interior solution to exist, the objective function Φ must be locally concave and its gradient must vanish. The first-order necessary conditions derived from this program provide a pair of elegant equations that encapsulate the economic logic of the optimal design.

Proposition 3.1 (Interior First-Order Characterization)**.** *Suppose F is C^1 with density $f > 0$ and b is C^1, increasing, and concave. If $(\varphi^\star, \lambda)$ is an interior maximizer of Φ, then it satisfies*

$$\boxed{\varphi^\star = b'(\lambda)} \qquad \textit{and} \qquad \boxed{\frac{b(\lambda)}{\lambda} - \varphi^\star = \frac{F(\varphi^\star)}{f(\varphi^\star)}}. \tag{7}$$

Conversely, any interior pair $(\varphi^\star, \lambda)$ that solves (7) and satisfies the second-order conditions is a local maximizer.

Proof. Differentiate Φ with respect to each choice variable. For $\lambda \in (0,1)$ and $\varphi^\star \in (\underline{\varphi}, \bar{\varphi})$,

$$\frac{\partial \Phi}{\partial \lambda} = F(\varphi^\star)\big(b'(\lambda) - \varphi^\star\big), \qquad \frac{\partial \Phi}{\partial \varphi^\star} = f(\varphi^\star)\big(b(\lambda) - \varphi^\star\lambda\big) - F(\varphi^\star)\lambda.$$

Interior optimality requires both derivatives to be zero. The first yields $\varphi^\star = b'(\lambda)$. Substituting into the second gives

$$f(\varphi^\star)\Big(b(\lambda) - \varphi^\star\lambda\Big) = F(\varphi^\star)\lambda \quad \Longleftrightarrow \quad \frac{b(\lambda)}{\lambda} - \varphi^\star = \frac{F(\varphi^\star)}{f(\varphi^\star)}.$$

The converse statement is standard: if (7) holds and the Hessian is negative semidefinite at $(\varphi^\star, \lambda)$, the point is a local maximizer. □

The two conditions in (7) are the central result of our positive analysis, and each has a deep economic interpretation that connects our specific model to broader principles of mechanism design and industrial organization.

The first equation, $\varphi^\star = b'(\lambda)$, can be termed the **Optimal Hazard Rule**. It governs the reliability dimension of the contract. The left-hand side, $\varphi^\star$, is the disutility from interruption for the marginal buyer, the most sensitive customer who is just indifferent between the preemptible and guaranteed options. The right-hand side, $b'(\lambda)$, is the marginal salvage value, representing the incremental benefit to the provider from a small increase in the overall hazard rate. Drawing on our microfoundation in Section 2.4, $b'(\lambda)$ is precisely the shadow value of capacity, $\kappa(\lambda)$, in the marginal state of congestion where the next interruption would be triggered. This condition thus creates a powerful alignment between the market-facing and operational aspects of the provider's strategy. It dictates that the hazard should be increased to the exact point where the marginal operational gain from freeing up capacity equals the marginal harm experienced by the most vulnerable customer in the preemptible pool.

The second equation, $\frac{b(\lambda)}{\lambda} - \varphi^\star = \frac{F(\varphi^\star)}{f(\varphi^\star)}$, is the **Optimal Cutoff Rule**. It governs the market segmentation dimension. The left-hand side represents the provider's net marginal profit from expanding the preemptible market. The term $\frac{b(\lambda)}{\lambda}$ is the average salvage value per unit of hazard, while $\varphi^\star$ is related to the price discount required to keep the marginal buyer. Their difference captures the incremental surplus the provider extracts by enrolling the marginal type $\varphi^\star$. The right-hand side, $\frac{F(\varphi^\star)}{f(\varphi^\star)}$, is the inverse hazard rate of the type distribution. This term is the canonical representation of information rents in screening models in the style of Mussa and Rosen (1978). It reflects the marginal cost of screening: to persuade the marginal buyer $\varphi^\star$ to select the preemptible option, the provider must offer a sufficiently large discount, and this discount benefits all infra-marginal types $\varphi < \varphi^\star$ as well. The provider thus expands the preemptible market (increases $\varphi^\star$) until the marginal gain from selling to one more buyer equals this marginal screening cost.

For a well-behaved interior optimum to exist, we must also consider the second-order conditions. It is helpful to record the curvature and cross-partial at an interior candidate, both for intuition and for the second-order conditions. Simple differentiation yields

$$\frac{\partial^2 \Phi}{\partial \lambda^2} \;=\; F(\varphi^\star)\, b''(\lambda) \;\le 0, \qquad \frac{\partial^2 \Phi}{\partial \varphi^\star \partial \lambda} \;=\; f(\varphi^\star)\big(b'(\lambda) - \varphi^\star\big) - F(\varphi^\star).$$

The own-curvature in λ inherits the concavity of b, so $\partial^2\Phi/\partial\lambda^2 \le 0$ is satisfied. At any point satisfying the first-order conditions (7), the cross-partial simplifies to $\frac{\partial^2 \Phi}{\partial \varphi^\star \partial \lambda} = -F(\varphi^\star) < 0$. This indicates that the hazard λ and the cutoff $\varphi^\star$ are locally strategic substitutes from the provider's perspective, which helps stabilize the optimum. The own-curvature in $\varphi^\star$ can be written, using (7), as

$$\frac{\partial^2 \Phi}{\partial \varphi^{\star\,2}} = f'(\varphi^\star)\frac{F(\varphi^\star)}{f(\varphi^\star)}\,\lambda - 2f(\varphi^\star)\lambda,$$

whose sign is generally ambiguous without additional shape restrictions on f. This is not a deficiency of the model but a standard feature of screening problems. Rather than assuming global concavity of Φ in $(\varphi^\star, \lambda)$, our analysis will rely on global arguments, including a careful check of the boundaries, and on the structure imposed by (7).

Two remarks clarify existence and the role of regularity. First, let $A(\varphi) \equiv F(\varphi)/f(\varphi)$ denote the inverse hazard ratio. Under the usual *regularity* assumption that A is increasing on $[\underline{\varphi}, \bar{\varphi}]$ (a condition satisfied by many common distributions, including those with log-concave densities), the second equation in (7) can be viewed as a single-crossing condition in $\varphi^\star$ for each fixed λ. This delivers a unique $\varphi^\star(\lambda)$ solving the second equation whenever $b(\lambda)/\lambda$ lies in the range of $\varphi \mapsto \varphi + A(\varphi)$. Second, the the strict concavity of b ensures that $\lambda \mapsto b'(\lambda)$ is strictly decreasing; hence the first equation selects at most one λ for any given $\varphi^\star$. Together, these monotonicities are crucial for ensuring a well-behaved solution.

While the first-order conditions are necessary, they do not guarantee that an interior solution exists for all possible parameterizations. It is useful to have a more direct test. The following lemma provides a convenient set of sufficient conditions on the primitives for an interior solution to exist.

Lemma 3.1 (A Convenient Interiority Test)**.** *Assume b is C^1, strictly increasing and strictly concave with $b(0) = 0$, and $A(\varphi) = F(\varphi)/f(\varphi)$ is continuous and finite on $[\underline{\varphi}, \bar{\varphi}]$. If*

$$A\big(b'(0^+)\big) \;>\; 0 \qquad \textit{and} \qquad \frac{b(1)}{1} - b'(1) \;>\; A\big(b'(1)\big), \tag{8}$$

then there exists $(\varphi^\star, \lambda) \in (\underline{\varphi}, \bar{\varphi}) \times (0, 1)$ that solves (7).

Proof. Define $H(\lambda) \equiv \frac{b(\lambda)}{\lambda} - b'(\lambda) - A(b'(\lambda))$ for $\lambda \in (0, 1]$, and extend by continuity at 0 using

$\lim_{\lambda \downarrow 0} b(\lambda)/\lambda = b'(0^+)$. Concavity gives $b(\lambda)/\lambda \geq b'(\lambda)$, with equality only at $\lambda = 0$. Hence $\lim_{\lambda \downarrow 0} H(\lambda) = 0 - A(b'(0^+)) < 0$ under the first inequality in (8). Continuity of H follows from continuity of b' and A. At $\lambda = 1$, $H(1) = \frac{b(1)}{1} - b'(1) - A(b'(1)) > 0$ by the second inequality. By the intermediate value theorem there exists $\lambda \in (0, 1)$ with $H(\lambda) = 0$. Setting $\varphi^\star = b'(\lambda)$ then satisfies both equations in (7). Since b' maps $(0, 1)$ into $(b'(1), b'(0^+))$ and b is strictly increasing, $\varphi^\star$ lies in $(\underline{\varphi}, \bar{\varphi})$ whenever $[b'(1), b'(0^+)] \subseteq (\underline{\varphi}, \bar{\varphi})$; otherwise, replace $[\underline{\varphi}, \bar{\varphi}]$ by the effective overlap interval. □

The test in (8) is easy to check in practice and has a clear economic meaning. The left inequality, $A(b'(0^+)) > 0$, requires that there is some positive mass of buyers with low interruption costs and that the initial marginal salvage is valuable enough to fall within the support of types. The right inequality compares the *average-minus-marginal* salvage at full hazard, $b(1) - b'(1)$, to the screening markup at the lower endpoint of the salvage values, $A(b'(1))$. It is satisfied whenever the salvage function is sufficiently "curvy" and the tail of the type distribution is not too heavy, ensuring that the trade-offs are balanced enough to warrant an interior solution rather than one at the boundary.

3.2 Boundaries, Existence, and When Corners are Optimal

The characterization of the interior optimum provides the core economic logic, but a complete analysis requires examining the boundaries of the choice set. The provider's problem is the maximization of a continuous function, $\Phi(\varphi^\star, \lambda)$, over a compact set, $[\underline{\varphi}, \bar{\varphi}] \times [0, 1]$. The Weierstrass Extreme Value Theorem therefore guarantees that a maximum always exists. If the conditions for an interior solution are not met, this maximum must lie on one of the four boundaries: $\lambda = 0$, $\lambda = 1$, $\varphi^\star = \underline{\varphi}$, or $\varphi^\star = \bar{\varphi}$. These boundary cases are not mere technicalities; they correspond to distinct and economically meaningful market outcomes.

We begin by providing conditions under which the provider will choose an extreme hazard level, either abandoning the preemptible option entirely or making it maximally risky. Let $c_0 \equiv b'(0^+)$ denote the marginal value of the very first unit of reclaimed capacity and $c_1 \equiv b'(1^-)$ denote the marginal value of the last.

Proposition 3.2 (Extreme Hazards)**.** *Assume b is C^1 on $(0, 1)$, increasing and concave with $b(0) = 0$, and F is continuous with density $f > 0$ on $[\underline{\varphi}, \bar{\varphi}]$.*

1. *(*No Preemption is Optimal*) If the marginal value of even the most valuable reclaimed capacity is less than the disutility of the least sensitive buyer, $c_0 \leq \underline{\varphi}$, then every maximizer satisfies $\lambda^\star = 0$. Moreover, for any $\varphi^\star \in [\underline{\varphi}, \bar{\varphi}]$, $\Phi(\varphi^\star, 0) = 0$, so the menu with $\lambda = 0$ delivers no incremental profit over the guaranteed product.*

2. *(*Maximal Preemption is Optimal*) If the marginal value of even the least valuable reclaimed capacity exceeds the disutility of the most sensitive buyer, $c_1 \geq \bar{\varphi}$, then every maximizer satisfies $\lambda^\star = 1$.*

Proof. Note that $\partial\Phi/\partial\lambda = F(\varphi^\star)\big(b'(\lambda) - \varphi^\star\big)$ for $\lambda \in (0,1)$.

(1) Fix $\varphi^\star$. Concavity of b implies $b'(\lambda) \leq b'(0^+) = c_0$ for all $\lambda \in [0,1]$, hence

$$\frac{\partial\Phi}{\partial\lambda}(\varphi^\star,\lambda) \;\leq\; F(\varphi^\star)\big(c_0 - \varphi^\star\big) \;\leq\; F(\varphi^\star)\big(\underline{\varphi} - \varphi^\star\big) \;\leq\; 0.$$

Thus $\Phi(\varphi^\star,\lambda)$ is weakly decreasing in λ for every $\varphi^\star$, and is maximized at $\lambda = 0$. Since $b(0) = 0$, $\Phi(\varphi^\star, 0) = F(\varphi^\star)\cdot 0 = 0$.

(2) Similarly, concavity implies $b'(\lambda) \geq b'(1^-) = c_1$ for all $\lambda \in [0,1]$. If $c_1 \geq \bar{\varphi}$ then

$$\frac{\partial\Phi}{\partial\lambda}(\varphi^\star,\lambda) \;\geq\; F(\varphi^\star)\big(\bar{\varphi} - \varphi^\star\big) \;\geq\; 0,$$

so Φ is weakly increasing in λ for every $\varphi^\star$, and is maximized at $\lambda = 1$. □

Proposition 3.2 formalizes the intuition that the interruption option is only valuable if the operational gains from reclaiming capacity are commensurate with the interruption costs imposed on customers. When the entire range of marginal salvage values lies outside the entire range of customer disutilities, the optimal hazard is pushed to a corner. In the first case, any positive hazard would impose a cost on the marginal buyer ($\varphi^\star \geq \underline{\varphi}$) that exceeds the benefit ($b'(\lambda) \leq c_0$), making interruptions unprofitable at any level. In the second case, every potential interruption is so valuable that the provider will increase the hazard to its physical maximum.

Even when the hazard is interior, the optimal cutoff may not be. To systematically find the global optimum, it is useful to characterize the optimal cutoff $\varphi^\star(\lambda)$ for any given hazard level λ. This reduces the two-dimensional search to a one-dimensional one. For this purpose, we define a helper function, $H_\lambda(\varphi)$, which represents the scaled derivative of the profit function with respect to $\varphi^\star$:

$$H_\lambda(\varphi) \;\equiv\; \frac{b(\lambda)}{\lambda} - \varphi - \frac{F(\varphi)}{f(\varphi)}.$$

The sign of $H_\lambda(\varphi)$ determines the direction of the provider's incentive to change the cutoff.

Lemma 3.2 (Cutoff Selection at a Fixed Hazard)**.** *Fix $\lambda \in (0,1]$ and assume $A(\varphi) \equiv F(\varphi)/f(\varphi)$ is continuous and strictly increasing on $[\underline{\varphi}, \bar{\varphi}]$ (regularity). Then:*

1. *H_λ is continuous and strictly decreasing in φ.*
2. *If $H_\lambda(\underline{\varphi}) \leq 0$, then $\Phi(\cdot,\lambda)$ is (weakly) decreasing in φ and the optimal cutoff at this λ is $\varphi^\star(\lambda) = \underline{\varphi}$.*

3. *If $H_\lambda(\bar{\varphi}) \geq 0$, then $\Phi(\cdot, \lambda)$ is (weakly) increasing in φ and the optimal cutoff at this λ is $\varphi^\star(\lambda) = \bar{\varphi}$.*

4. *If $H_\lambda(\underline{\varphi}) > 0 > H_\lambda(\bar{\varphi})$, there exists a unique interior $\varphi^\star(\lambda) \in (\underline{\varphi}, \bar{\varphi})$ satisfying $H_\lambda(\varphi^\star(\lambda)) = 0$, i.e.,*

$$\frac{b(\lambda)}{\lambda} - \varphi^\star(\lambda) = \frac{F(\varphi^\star(\lambda))}{f(\varphi^\star(\lambda))}.$$

Proof. By definition, H_λ is continuous. Since A is strictly increasing, $H'_\lambda(\varphi) = -1 - A'(\varphi) < 0$, proving (1). For $\lambda > 0$,

$$\frac{\partial \Phi}{\partial \varphi^\star}(\varphi, \lambda) = f(\varphi)\lambda\, H_\lambda(\varphi).$$

Thus the sign of H_λ determines the monotonicity of $\Phi(\cdot, \lambda)$, delivering (2) and (3). By the intermediate value theorem and strict monotonicity of H_λ, (4) follows. □

This lemma provides a practical recipe for finding the global optimum: for each candidate hazard λ, we can find the corresponding optimal $\varphi^\star(\lambda)$. By substituting this back into the objective function, we can define a concentrated profit function $\widehat{\Phi}(\lambda) = \Phi(\varphi^\star(\lambda), \lambda)$ and solve the full problem by maximizing this single-variable function over $\lambda \in [0, 1]$.

A crucial case that illustrates the importance of boundary solutions is when the salvage function is linear, $b(\lambda) = \beta\lambda$. As we reasoned before, this forces the hazard to an extreme.

Proposition 3.3 (Linear Salvage ⇒ Extreme Hazards)**.** *Suppose $b(\lambda) = \beta\lambda$ with $\beta \geq 0$ and F is continuous with $f > 0$ on $[\underline{\varphi}, \bar{\varphi}]$. Then every maximizer has $\lambda^\star \in \{0, 1\}$. In fact,*

$$\lambda^\star = \begin{cases} 0, & \text{if } \beta \leq \underline{\varphi}, \\ 1, & \text{if } \beta > \underline{\varphi}, \end{cases}$$

and, when $\lambda^\star = 1$, the optimal cutoff solves

$$\varphi^\star \in \arg\max_{\varphi \in [\underline{\varphi},\, \min\{\beta, \bar{\varphi}\}]} g(\varphi), \qquad g(\varphi) \equiv F(\varphi)\,(\beta - \varphi), \tag{9}$$

which is unique under regularity (increasing F/f) and satisfies $\beta - \varphi^\star = \frac{F(\varphi^\star)}{f(\varphi^\star)}$ if interior.

Proof. Here $\Phi(\varphi^\star, \lambda) = \lambda\, F(\varphi^\star)\,(\beta - \varphi^\star)$ is linear in λ for each $\varphi^\star$. If $\beta \leq \underline{\varphi}$, then $\beta - \varphi^\star \leq 0$ for all $\varphi^\star$, so Φ is weakly decreasing in λ and $\lambda^\star = 0$ is optimal. If $\beta > \underline{\varphi}$, pick any φ with $\underline{\varphi} \leq \varphi < \min\{\beta, \bar{\varphi}\}$; then $F(\varphi) > 0$ and $(\beta - \varphi) > 0$, so increasing λ raises Φ. Hence $\lambda^\star = 1$. Conditional on $\lambda = 1$, maximizing Φ over $\varphi^\star$ is (9). The derivative $g'(\varphi) = f(\varphi)(\beta - \varphi) - F(\varphi)$ vanishes iff $\beta - \varphi = F(\varphi)/f(\varphi)$; under regularity, the left side decreases in φ and the right side

increases, implying a unique interior maximizer when it lies in $(\underline{\varphi}, \min\{\beta, \bar{\varphi}\})$. Otherwise the maximizer is at the boundary $\varphi^\star = \min\{\beta, \bar{\varphi}\}$. □

This result starkly contrasts with the rich interior solution characterized by Proposition 3.1. It underscores that the concavity of the salvage function, arising from the provider's ability to surgically target interruptions to the most congested states, is the essential economic force that supports a nuanced, non-extreme reliability policy. Without it, the provider's choice becomes a blunt, all-or-nothing decision.

Finally, to complete our toolkit for finding the global optimum, we explicitly characterize the provider's problem along the cutoff boundaries.

Lemma 3.3 (Optimizing along $\varphi^\star = \underline{\varphi}$ and $\varphi^\star = \bar{\varphi}$)**.** *Assume F is continuous with $f > 0$ on $[\underline{\varphi}, \bar{\varphi}]$ and b is increasing and concave.*

1. *Along $\varphi^\star = \underline{\varphi}$,*

$$\max_{\lambda\in[0,1]} \Phi(\underline{\varphi}, \lambda) = F(\underline{\varphi}) \cdot \max_{\lambda\in[0,1]} \left\{ b(\lambda) - \underline{\varphi}\lambda \right\}.$$

   If $F(\underline{\varphi}) = 0$ (the continuous-support case), this edge yields zero incremental profit for all λ.

2. *Along $\varphi^\star = \bar{\varphi}$,*

$$\max_{\lambda\in[0,1]} \Phi(\bar{\varphi}, \lambda) = \max_{\lambda\in[0,1]} \left\{ b(\lambda) - \bar{\varphi}\lambda \right\},$$

   whose maximizers satisfy $b'(\lambda) \in [\bar{\varphi}^-, \bar{\varphi}^+]$; with concave b this reduces to the rule

$$\lambda^\sharp = \begin{cases} 0, & c_0 \le \bar{\varphi}, \\ 1, & c_1 \ge \bar{\varphi}, \\ (b')^{-1}(\bar{\varphi}), & \textit{otherwise}, \end{cases}$$

   where $(b')^{-1}$ is well defined by strict concavity.

Proof. (1) and (2) are single-variable concave maximization problems. For (1), the derivative w.r.t. λ is $F(\underline{\varphi})\big(b'(\lambda) - \underline{\varphi}\big)$, so any maximizer satisfies $b'(\lambda) \in [\underline{\varphi}^-, \underline{\varphi}^+]$; if $F(\underline{\varphi}) = 0$ the objective is flat and equals zero. For (2), the derivative is $b'(\lambda) - \bar{\varphi}$ and the stated conditions follow from concavity of b. □

These lemmas provide a complete map of the provider's optimization landscape. The global optimum is found by identifying the interior candidate from Proposition 3.1 and comparing its profitability against the best possible outcomes on each of the four boundaries, as characterized by Propositions 3.2 and Lemma 3.3. The next subsection will apply this machinery to derive concrete solutions for specific functional forms.

3.3 Canonical Families and Closed Forms

To derive clearer intuition and illustrate the mechanics of the optimal design, this subsection applies the general characterization from §3.1 and §3.2 to tractable parametric families for the type distribution and the salvage function. We focus on the benchmark case where buyer types are uniformly distributed, which yields a particularly clean structure for the screening problem. We then analyze the optimal menu under three distinct forms for the salvage technology: linear, exponential, and power-law. These cases highlight how the curvature of the salvage function is the primary determinant of the provider's reliability strategy.

Setup: Uniformly Distributed Types. Throughout this subsection, we assume that interruption sensitivities are uniformly distributed on the interval $[0, \bar{\varphi}]$. This implies a linear CDF, $F(\varphi) = \varphi/\bar{\varphi}$, and a constant density, $f(\varphi) = 1/\bar{\varphi}$. Consequently, the inverse hazard rate, a key component of the screening problem, takes a simple linear form:

$$A(\varphi) = \frac{F(\varphi)}{f(\varphi)} = \varphi.$$

Under this assumption, the second first-order condition from Proposition 3.1, which governs the optimal cutoff, simplifies dramatically to:

$$\frac{b(\lambda)}{\lambda} - \varphi^\star = \varphi^\star \quad \Longrightarrow \quad \frac{b(\lambda)}{\lambda} = 2\varphi^\star. \tag{10}$$

When combined with the first condition, $\varphi^\star = b'(\lambda)$, we obtain a single, powerful scalar identity that any interior optimal hazard λ must satisfy:

$$b(\lambda) = 2\lambda b'(\lambda). \tag{11}$$

This equation elegantly states that at the optimum, the total salvage value must be equal to twice the product of the hazard and the marginal salvage value. It provides a direct link between the global and local properties of the salvage function, and its solvability will determine whether an interior solution exists.

Uniform Types with Linear Salvage

We begin with the simplest case, a linear salvage function $b(\lambda) = \beta\lambda$, where $\beta \geq 0$ is the constant marginal value of reclaimed capacity. As established in Proposition 3.3, the linearity of the objective in λ forces the optimal hazard to an extreme.

Proposition 3.4 (Uniform Types, Linear Salvage)**.** *Suppose $\varphi \sim U[0, \bar{\varphi}]$ and $b(\lambda) = \beta\lambda$. Then every maximizer has $\lambda^\star \in \{0, 1\}$. If $\beta = 0$, $\lambda^\star = 0$ and the preemptible option collapses. If $\beta > 0$,*

$\lambda^\star = 1$ *and*

$$\varphi^\star \;=\; \min\left\{\frac{\beta}{2},\, \bar{\varphi}\right\}, \qquad \Pi - v \;=\; \begin{cases} \dfrac{\beta^2}{4\bar{\varphi}}, & \beta \le 2\bar{\varphi}, \\ \beta - \bar{\varphi}, & \beta > 2\bar{\varphi}. \end{cases}$$

Proof. The profit function $\Phi(\varphi^\star, \lambda) = \lambda(\varphi^\star/\bar{\varphi})(\beta - \varphi^\star)$ is linear in λ. If $\beta = 0$, $\Phi \equiv 0$, so we take $\lambda^\star = 0$. If $\beta > 0$, the provider can always choose a $\varphi^\star < \beta$, making the term $(\beta - \varphi^\star)$ positive and thus making it optimal to set $\lambda^\star = 1$. Conditional on $\lambda = 1$, the provider chooses $\varphi^\star$ to maximize the concave quadratic $(\varphi/\bar{\varphi})(\beta - \varphi)$ over $\varphi \in [0, \bar{\varphi}]$. The unconstrained maximizer is $\beta/2$. Truncating this at the upper bound $\bar{\varphi}$ gives the stated optimal cutoff. The profit follows from substituting $\varphi^\star$ back into the objective. □

This result confirms that without diminishing returns to salvage, the provider's reliability choice is binary. The cutoff $\varphi^\star = \beta/2$ represents the classic trade-off between price $(\beta - \varphi^\star)$ and quantity $(F(\varphi^\star))$ for a monopolist facing a linear demand curve.

Uniform Types with Exponential Salvage

A more realistic salvage technology exhibits diminishing returns. A canonical representation is the exponential function $b(\lambda) = \beta\big(1 - e^{-a\lambda}\big)$, where $\beta > 0$ is the maximum potential salvage and $a > 0$ is a curvature parameter indicating how quickly marginal returns diminish. Substituting this into our characteristic equation for an interior optimum, $b(\lambda) = 2\lambda b'(\lambda)$, yields:

$$\beta\big(1 - e^{-a\lambda}\big) \;=\; 2\lambda \cdot \beta a e^{-a\lambda} \quad\Longleftrightarrow\quad e^{a\lambda} \;=\; 1 + 2a\lambda.$$

Letting $x \equiv a\lambda$, this becomes $e^x = 1 + 2x$. Besides the trivial root $x = 0$, this equation has a unique positive solution, which can be expressed using the Lambert W function:

$$x^\star \;=\; -W_{-1}\left(-\frac{e^{-1/2}}{2}\right) - \frac{1}{2} \;\approx\; 1.25643. \tag{12}$$

This gives a unique candidate for the interior optimal hazard, $\lambda^\circ = x^\star/a$, and a corresponding cutoff, $\varphi^{\star\circ} = b'(\lambda^\circ)$.

Proposition 3.5 (Uniform Types, Exponential Salvage)**.** *Let $\varphi \sim U[0, \bar{\varphi}]$ and $b(\lambda) = \beta(1 - e^{-a\lambda})$ with $\beta, a > 0$. Define $x^\star$ by (12) and the candidate $(\lambda^\circ, \varphi^{\star\circ})$ by $\lambda^\circ = x^\star/a$ and $\varphi^{\star\circ} = b'(\lambda^\circ)$. Then the global maximizer is:*

1. ***Interior Solution*** *if the candidate is feasible ($a \ge x^\star$ and $\varphi^{\star\circ} \le \bar{\varphi}$): $\lambda^\star = \lambda^\circ$ and $\varphi^\star = \varphi^{\star\circ}$. The incremental profit is $\Pi - v = \frac{x^\star}{(1+2x^\star)^2} \cdot \frac{\beta^2 a}{\bar{\varphi}}$.*
2. ***Boundary Solution*** *otherwise. The optimum lies either at the hazard boundary $\lambda = 1$ or the cutoff boundary $\varphi^\star = \bar{\varphi}$, determined by the conditions in Propositions 3.2 and Lemma 3.3.*

The interior solution provides clear comparative statics. The optimal hazard $\lambda^\star = x^\star/a$ is inversely proportional to the curvature a: when diminishing returns set in more quickly, the provider interrupts less. The optimal cutoff $\varphi^\star$ increases with the salvage scale β and curvature a. This case demonstrates that sufficient curvature in the salvage technology is essential for sustaining a finely tuned, interior reliability policy.

Uniform Types with Power-Law Salvage

As a final example, consider the power-law family, $b(\lambda) = \beta\lambda^\gamma$ with $\gamma \in (0, 1]$. Applying the interior solution condition $b(\lambda) = 2\lambda b'(\lambda)$ gives:

$$\beta\lambda^\gamma \;=\; 2\lambda \cdot (\beta\gamma\lambda^{\gamma-1}) \quad\Longleftrightarrow\quad 1 \;=\; 2\gamma.$$

This yields a surprising and instructive result: an interior solution can exist *only* for the edge case where $\gamma = 1/2$.

Proposition 3.6 (Power Salvage and the Square-Root Knife Edge)**.** *Let $\varphi \sim U[0, \bar{\varphi}]$ and $b(\lambda) = \beta\lambda^\gamma$.*

1. *If $\gamma = 1/2$, the condition $b(\lambda) = 2\lambda b'(\lambda)$ is satisfied for all $\lambda > 0$. The provider is indifferent over a continuum of pairs $(\lambda, \varphi^\star)$ satisfying $\varphi^\star = b'(\lambda) = \beta/(2\sqrt{\lambda})$, all of which yield the same profit $\Pi - v = \frac{\beta^2}{4\bar{\varphi}}$.*

2. *If $\gamma \neq 1/2$, no interior solution exists. The global optimum must lie on one of the boundaries of the choice space.*

The square-root case reveals a peculiar indeterminacy. The provider can achieve the same profit with a low-hazard, high-cutoff menu as with a high-hazard, low-cutoff menu. Any small perturbation away from $\gamma = 1/2$ breaks this indifference and pushes the solution to a corner. If $\gamma > 1/2$, the salvage function is "less concave" than square-root, and the provider prefers higher hazards, pushing the solution towards $\lambda = 1$. If $\gamma < 1/2$, the function is "more concave," and the provider prefers lower hazards.

Taken together, these canonical examples paint a clear picture. The existence and nature of an interior optimal menu are critically dependent on the curvature of the salvage function. Linear salvage leads to extreme policies. Strictly concave salvage, like the exponential family, supports a unique and stable interior design. Specific functional forms, like the power-law family, can exhibit edge properties. This underscores the importance of understanding the operational realities of capacity management, as they are the ultimate drivers of the menu structure we observe in the market.

3.4 Operational and Policy Constraints

The unconstrained optimization problem provides a crucial theoretical benchmark. In practice, however, providers often design their menus under additional constraints, either self-imposed for operational simplicity or externally imposed by regulators or consumer protection laws. Two such constraints are particularly relevant: limits on the maximum permissible hazard (*reliability caps*) and rules governing how prices and refunds can be displayed to consumers. These constraints can significantly alter the optimal menu and, importantly, can break the implementational neutrality of refunds.

Reliability Caps. A common form of service level agreement (SLA) or regulatory requirement is a cap on unreliability. We model this as a simple constraint on the maximum hazard:

$$\lambda \;\le\; \bar{\lambda}, \quad \text{for some } \bar{\lambda} \in (0, 1].$$

The provider's problem becomes a constrained maximization. If the unconstrained optimum $\lambda^\star$ is already below $\bar{\lambda}$, the cap is slack and has no effect. If $\lambda^\star > \bar{\lambda}$, the cap binds. In this case, the provider will set $\lambda = \bar{\lambda}$ and then choose the cutoff $\varphi^\star$ to maximize profit conditional on this fixed hazard. This one-dimensional optimization is solved by Lemma 3.2, yielding a cutoff $\varphi^\dagger(\bar{\lambda})$ that satisfies $\frac{b(\bar{\lambda})}{\bar{\lambda}} - \varphi^\dagger(\bar{\lambda}) = A(\varphi^\dagger(\bar{\lambda}))$.

Proposition 3.7 (When the Reliability Cap Binds)**.** *A reliability cap $\bar{\lambda}$ binds if and only if the unconstrained optimal hazard $\lambda^\star$ exceeds $\bar{\lambda}$. When it binds, the constrained optimum is $(\lambda = \bar{\lambda}, \varphi^\star = \varphi^\dagger(\bar{\lambda}))$. At this point, the marginal value of salvage exceeds the marginal buyer's disutility: $b'(\bar{\lambda}) > \varphi^\dagger(\bar{\lambda})$.*

The final inequality is the shadow price of the constraint: because the provider is prevented from increasing the hazard to its desired level, it is left in a state of "under-preemption," where the marginal benefit of another interruption is strictly positive.

Refund Floors and Price-Display Restrictions. Our analysis so far has treated refunds as purely implementational, a consequence of the provider's ability to freely adjust prices. However, platform policies or regulations often impose constraints on pricing, which makes the choice of refund r strategically important. Common rules include a floor on refunds ($r \ge \underline{r}$), a floor on the advertised price ($p_P \ge \underline{p}$), or a cap on the advertised discount ($p_G - p_P \le \bar{\Delta}$).

These rules break the simple mapping from $(\varphi^\star, \lambda)$ to profits because not all pairs are implementable. Using the indifference condition $p_G - p_P = \lambda(\varphi^\star - r)$ with $p_G = v$, these constraints can be translated into a single feasibility constraint on the choice variables:

$$\lambda\big(\varphi^\star - \underline{r}\big) \le \chi, \tag{13}$$

where $\chi \equiv \min\{v - \underline{p}, \bar{\Delta}\}$ is a parameter capturing the stringency of the display rule. This inequality defines a feasible region in the $(\varphi^\star, \lambda)$ space, as shown in Figure **??**.

Proposition 3.8 (Effects of Binding Display Constraints)**.** *Suppose a display constraint of the form in* (13) *binds. Then, relative to the unconstrained optimum, the provider chooses a menu with both a* ***lower hazard*** *and a* ***lower cutoff****.*

Intuitively, a binding refund floor-plus-display rule forbids deep *posted* discounts despite the operator's desire to screen aggressively when salvage is valuable. The platform responds by lowering both the hazard and the cutoff until the net advertised discount saturates the bound (χ). Figure 3 illustrates this distortion graphically. The display constraint carves out a feasible region in the $(\lambda, \varphi^\star)$ space. If the unconstrained profit-maximizing point lies outside this region, the provider must retreat to a new, constrained optimum on the feasibility boundary. As the figure shows, this forced move results in an allocation with both a lower hazard and a smaller preemptible market (a lower cutoff) than the provider would choose in an unconstrained environment. This demonstrates how consumer-facing rules, while seemingly simple, can fundamentally alter the provider's screening strategy and the resulting market structure

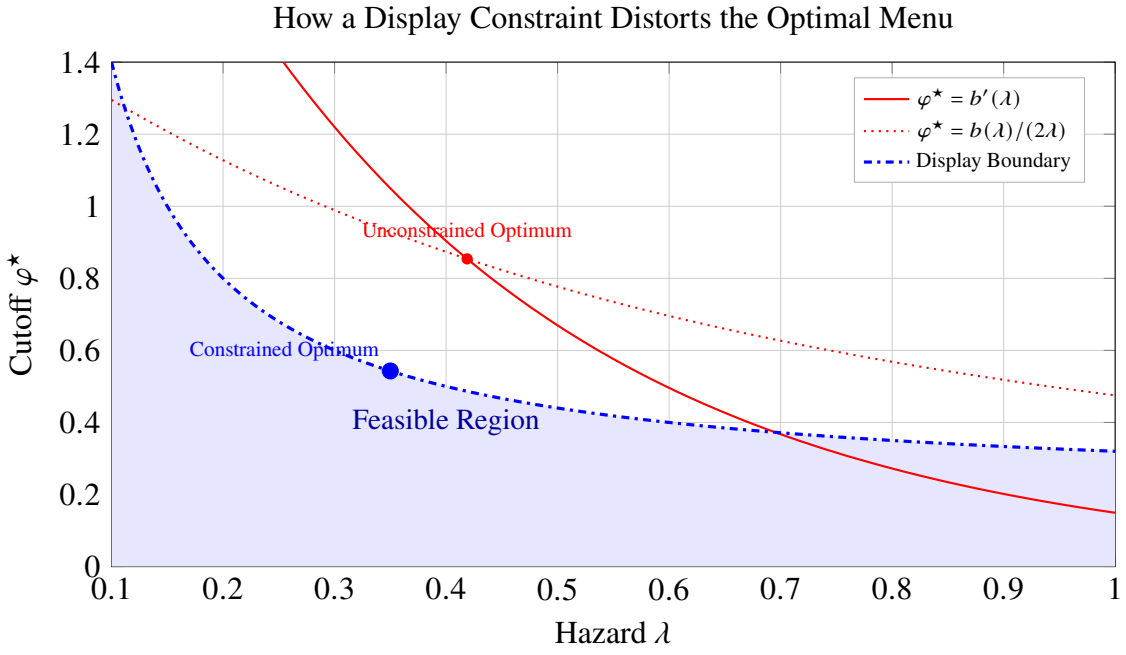


Figure 3: Illustration of a binding display constraint. The unconstrained optimum is determined by the intersection of the two first-order condition curves. Since this point is infeasible, the provider chooses a constrained optimum on the boundary, featuring a lower hazard and cutoff.

From a welfare perspective, both constraints limit the operator's ability to align marginal salvage with marginal disutility at the cutoff; the cap pushes the system toward under-preemption

$(b'(\lambda) > \varphi^\star)$, while the display rule depresses the cutoff below the second-best screen. In Section 5, we will quantify these wedges and characterize when such rules can nonetheless raise total surplus.

When both a reliability cap and a display rule are present, the provider's feasible set is the intersection of the regions defined by each constraint, which can lead to complex interactions. Figure 4 illustrates such a scenario. The unconstrained optimum lies in a region that violates both policies. The provider is forced to retreat to the boundary of the feasible set. In this example, the new optimum lies at the "kink" where both the reliability cap and the display rule bind simultaneously. The resulting allocation is a compromise shaped by both policies, exhibiting the under-preemption characteristic of a binding reliability cap $(b'(\bar{\lambda}) > \varphi^\star)$ while also being constrained by the display rule's limit on the net discount. This demonstrates how layers of simple regulations can interact to significantly shape the final market outcome.

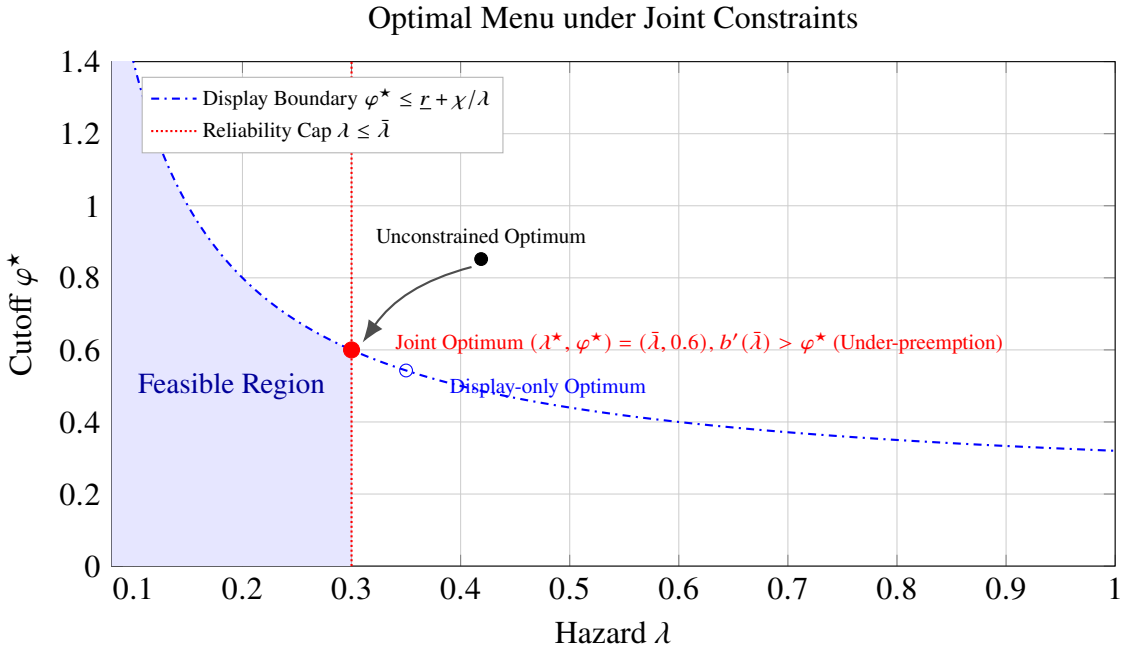


Figure 4: Interaction of a reliability cap and a display rule. The feasible set is the intersection of the regions defined by each constraint. When the unconstrained optimum is infeasible, the provider is pushed to the boundary. In this illustration, the optimum occurs at the kink where both constraints bind, resulting in an allocation that is distorted relative to the unconstrained ideal.

4 Equivalences and Implementations

The kill-option contract, while developed to capture the specific institutional features of modern cloud service markets, does not exist in a theoretical vacuum. Its underlying economic structure resonates deeply with some of the most foundational concepts in the economics of screening, resource allocation, and industrial organization. This section is dedicated to making these connections

precise by establishing formal equivalences between our model and two canonical mechanisms: the theory of damaged goods and the theory of priority auctions.

These equivalences serve a dual purpose. First, they provide a powerful conceptual framework for understanding the economic nature of preemptible instances. By mapping our model onto these well-understood structures, we can import a rich set of intuitions and interpret the provider's strategy in a broader theoretical context. Second, and perhaps more importantly, these connections enrich the classical theories themselves. They demonstrate how the abstract concepts of "quality degradation" and "priority" can be implemented in a dynamic, operational setting where the key attributes of the service are not fixed but are instead endogenously determined by state-contingent policies. This analysis bridges the often-separate literatures of marketing (on versioning and pricing) and operations (on congestion and resource allocation), showing them to be two sides of the same coin in the context of cloud services.

4.1 Damaged-Goods Equivalence

The theory of damaged goods, most famously articulated by Deneckere and McAfee (1996) provides a compelling explanation for why a monopolist might find it profitable to intentionally degrade a product. The logic is that by offering a damaged (and cheaper) version, the firm can segment the market, selling to price-sensitive consumers without cannibalizing sales of its premium product to high-willingness-to-pay consumers. This strategy is only profitable if the act of damaging the good allows for effective price discrimination and, crucially, is associated with a lower effective cost for the firm. We now demonstrate that our spot menu is a sophisticated, probabilistic incarnation of this very strategy.

To formalize this connection, we perform a simple re-parameterization of our model. Instead of framing the preemptible contract in terms of a "bad", the interruption hazard λ, we can frame it in terms of a "good", the level of *reliability*, which we define as $q \equiv 1 - \lambda$. In this quality space, the guaranteed contract corresponds to a top-quality product with $q_G = 1$, while the preemptible contract is a lower-quality version with $q_P = 1 - \lambda < 1$.

Furthermore, we can consolidate the pricing terms by defining an *effective price* for the preemptible product, $\tilde{p}_P$, which nets out the expected refund:

$$\tilde{p}_P \equiv p_P - r\lambda = p_P - r(1-q).$$

With these new variables, the buyer's utility and the seller's profit for the preemptible option can be rewritten in the canonical language of a vertical quality differentiation model. A buyer of type φ choosing the lower-quality option now has an expected utility of:

$$U_P(\varphi) = v - p_P - \varphi\lambda + r\lambda = v - (p_P - r\lambda) - \varphi\lambda = v - \tilde{p}_P - \varphi(1-q).$$

This is the utility of a consumer who values the baseline product at v but has a type-dependent disutility φ for each unit of "quality degradation" $1-q$. The seller's profit from this buyer becomes:

$$\pi_P = p_P + b(\lambda) - r\lambda = (p_P - r\lambda) + b(\lambda) = \tilde{p}_P + b(1-q).$$

This final expression reveals the core of the equivalence. The seller's profit is the effective price plus a term that depends on the quality level. We can interpret this as a model where producing the high-quality good ($q = 1$) has a normalized cost of zero, while producing a lower-quality good of level q generates a **cost saving** of $b(1-q)$. The salvage value from reclaiming capacity in our model plays the exact same economic role as the cost advantage from using cheaper materials or disabling features in a traditional damaged-goods setting.

Proposition 4.1 (Damaged-Goods Equivalence)**.** *Consider the posted menu* $(p_G, q_G = 1)$ *and* $(p_P, q \in [0,1])$ *under the primitives above. Then:*

(i) For any (p_G, p_P, q, r)*, the induced cutoff and demands depend only on* $(p_G, \tilde{p}_P, q)$*, with*

$$\varphi^\star = \frac{p_G - \tilde{p}_P}{1-q}, \qquad D_P = F(\varphi^\star), \quad D_G = 1 - F(\varphi^\star).$$

(ii) With $p_G^\star = v$*, the seller's program reduces to*

$$\max_{q\in[0,1],\ \varphi^\star\in[\underline{\varphi},\bar{\varphi}]} \Psi(\varphi^\star, q) \ \equiv\ F(\varphi^\star)\Big(b(1-q) - \varphi^\star(1-q)\Big), \tag{14}$$

which is identical to (5) under the change of variables $q = 1 - \lambda$*.*

(iii) If $(\varphi^\star, q)$ *is interior, the first-order conditions are*

$$\boxed{\varphi^\star = b'(1-q)} \qquad \text{and} \qquad \boxed{\frac{b(1-q)}{1-q} - \varphi^\star = \frac{F(\varphi^\star)}{f(\varphi^\star)}}, \tag{15}$$

i.e., marginal cost savings from a further degradation $dq < 0$ *equal the cutoff disutility, and the average net margin per preemptible buyer equals the inverse hazard ratio.*

Conversely, given any $(\tilde{p}_P, q)$ *that implements a cutoff* $\varphi^\star$*, there exists a continuum of* (p_P, r) *with* $p_P = \tilde{p}_P + r(1-q)$ *that implements the same allocation and profits in the original preemption model.*

Proof. (i) From (4), indifference between G and P is

$$v - p_G \ =\ v - \tilde{p}_P - \varphi^\star(1-q) \implies \varphi^\star = \frac{p_G - \tilde{p}_P}{1-q},$$

and the single-crossing property yields the cutoff demand split.

(ii) Aggregate profits are

$$\Pi \;=\; (1 - F(\varphi^\star))\, p_G + F(\varphi^\star)\big(\tilde{p}_P + b(1-q)\big).$$

Using $p_G = v$ (IR binds) and the indifference relation $\tilde{p}_P = p_G - \varphi^\star(1-q) = v - \varphi^\star(1-q)$, we get

$$\Pi \;=\; v + F(\varphi^\star)\Big(b(1-q) - \varphi^\star(1-q)\Big) \equiv v + \Psi(\varphi^\star, q).$$

This is (14) and is identical to (5) with $q = 1 - \lambda$.

(iii) Differentiate Ψ at an interior pair:

$$\frac{\partial \Psi}{\partial q} = F(\varphi^\star)\Big(-b'(1-q) + \varphi^\star\Big), \qquad \frac{\partial \Psi}{\partial \varphi^\star} = f(\varphi^\star)\big(b(1-q) - \varphi^\star(1-q)\big) - F(\varphi^\star)(1-q).$$

Setting both to zero yields (15). The implementability claim follows since only $\tilde{p}_P$ enters utilities and profits. □

This formal equivalence is powerful because it allows us to immediately import the canonical insights of the damaged-goods literature. Two corollaries are immediate.

Corollary 4.1 (No Degradation Without a Cost Advantage)**.** *If the salvage value is zero for all hazard levels, $b(\lambda) \equiv 0$, then it is never optimal to offer a preemptible contract. The unique optimum is $\lambda^\star = 0$ ($q^\star = 1$), and the menu collapses to a single, guaranteed product.*

This result is fundamental: damaging reliability is only profitable if it generates a real economic benefit for the provider. Without a valuable alternative use for reclaimed capacity, offering a risky product at a discount would only serve to cannibalize revenue from the premium product.

Corollary 4.2 (Profitable Degradation with Positive Marginal Salvage)**.** *If there is any positive marginal value to reclaiming capacity at zero hazard, $b'(0^+) > \underline{\varphi}$, and there is a positive mass of buyers with low interruption costs, then introducing a small amount of unreliability ($\lambda > 0$) is strictly more profitable than offering only a single guaranteed product.*

This corollary shows that the incentive to version is very general. As long as there is some heterogeneity in the market and some value to be gained from reclaiming capacity, the provider will always find it profitable to introduce a lower-quality, preemptible option. The spot contract is thus a natural outcome in any environment with scarce, re-allocatable resources and heterogeneous user sensitivities.

4.2 Equivalence to Priority Auctions

A second, equally fundamental equivalence connects our posted-price menu to the extensive literature on priority pricing and auctions for allocating scarce resources (Mendelson and Whang, 1990; Ha, 2001; Maglaras and Zeevi, 2005). In a congested system, a classic way to achieve an efficient allocation is to assign priority to users based on their cost of delay or interruption. This can be implemented via a mechanism, such as an auction, where users bid for their place in the service queue. We now show that our simple, two-price menu achieves the same allocative outcome as a more complex, state-contingent priority auction, providing a practical implementation of the auction's theoretical ideal.

Consider a direct revelation mechanism where buyers report their type $\hat{\varphi}$. Based on these reports, the provider implements a resource allocation rule with two components:

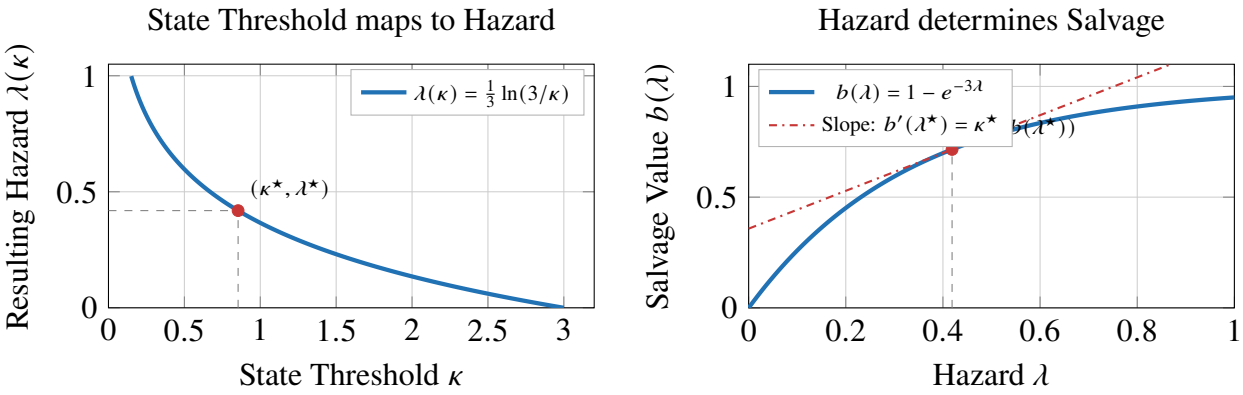
1. **Priority Assignment:** An assignment rule maps reports to one of two classes: a high-priority "guaranteed" class that is never interrupted, and a low-priority "preemptible" class that is subject to interruption. To be incentive compatible, this rule must be monotonic, assigning higher priority to those who report higher interruption costs. This implies a cutoff rule: buyers with $\hat{\varphi} \geq \varphi^\star$ get guaranteed service.

2. **Interruption Rule:** A state-contingent rule determines when the low-priority class is interrupted. As established in our salvage microfoundation, the optimal rule is a threshold policy: interrupt if and only if the shadow value of capacity $\mu(s)$ exceeds some state threshold κ.

The provider's problem is to choose the optimal pair of thresholds $(\varphi^\star, \kappa)$ and a set of transfers (prices and refunds) that make it incentive-compatible for buyers to report their types truthfully and that maximize profit. A cornerstone result from mechanism design is that in this single-parameter environment, the allocation and expected payoffs are pinned down by the envelope theorem, regardless of the specific format of the auction (e.g., VCG, uniform price).

Proposition 4.2 (Allocation and Payment Equivalence)**.** *Fix a state-contingent priority rule $h_\kappa(s) = \mathbf{1}\{\mu(s) \geq \kappa\}$. Suppose a truthful priority auction assigns guaranteed service if and only if $\varphi \geq \varphi^\star$. Then there exists a posted menu with the same hazard schedule, a guaranteed price $p_G = v$, and a preemptible price and refund schedule that implements the* same *cutoff, the* same *ex-ante utilities for every type, and hence the* same *expected profit. Conversely, any optimal posted menu is implementable by a corresponding truthful priority auction.*

Proof. By the envelope theorem, the utility of a type φ in a truthful priority auction with cutoff $\varphi^\star$ and hazard $\lambda(\kappa)$ must satisfy $U'(\varphi) = -\lambda(\kappa)$ for $\varphi < \varphi^\star$ and $U'(\varphi) = 0$ for $\varphi > \varphi^\star$. With the normalization that the marginal type $\varphi^\star$ earns zero surplus ($U(\varphi^\star) = 0$), this integrates to $U(\varphi) = \lambda(\kappa)(\varphi^\star - \varphi)$ for $\varphi < \varphi^\star$ and $U(\varphi) = 0$ otherwise. Our posted menu with $p_G = v$ and



(a) The state threshold κ determines the total hazard λ. A higher threshold means fewer states trigger an interruption, resulting in a lower hazard.

(b) The total hazard λ determines the expected salvage value $b(\lambda)$. The slope of the salvage function at any point is the marginal state threshold, $b'(\lambda) = \kappa$.

Figure 5: The Relationship Between State Threshold (κ), Hazard (λ), and Salvage ($b(\lambda)$)

$p_P = v - (\varphi^\star - r)\lambda$ produces exactly this utility schedule (see proof of Proposition 2.1). By the revenue equivalence principle, since the allocation (the cutoff $\varphi^\star$) and the utility function $U(\varphi)$ are identical across the two mechanisms, the provider's expected profit must also be identical. □

This equivalence is of great practical importance. It implies that a provider does not need to run a complex, continuous auction to achieve an efficient, state-contingent allocation of resources. A simple, transparent, and scalable posted menu with two options can achieve the same outcome. This is particularly valuable in markets like cloud computing where customers may value price stability and predictability, and where the cognitive costs of participating in continuous auctions could be a barrier to entry. The posted menu acts as a simple user interface that implements the sophisticated logic of a priority auction "under the hood."

The deepest connection comes from uniting the two mechanisms' optimality conditions:

Corollary 4.3 (Optimal Threshold Alignment)**.** *An optimal menu, characterized by $\varphi^\star = b'(\lambda)$, is equivalent to an optimal priority auction where the priority threshold ($\varphi^\star$) is set equal to the state-interruption threshold (κ), since $\kappa = b'(\lambda)$.*

This result provides a powerful and intuitive conclusion: the provider's optimal strategy aligns the market-facing decision of *who* to place at risk with the operational decision of *when* to reclaim capacity. The price-refund menu is calibrated so that the marginal buyer who self-selects into the preemptible pool has an interruption cost exactly equal to the value of capacity in the marginal state of congestion. The posted menu, therefore, is not just a blunt pricing tool; it is a precision instrument for implementing a sophisticated state-contingent resource allocation policy.

In sum, the "priority auction vs. posted menu" dichotomy is largely semantic in our baseline model. Both are simply alternative ways to implement the same fundamental screening logic,

governed by the same operational service interruption rule. The state threshold κ determines when capacity is reclaimed; the cutoff $\varphi^\star$ determines who bears the risk; and prices and refunds are the transfers that align incentives and distribute the surplus.

5 Welfare and Policy

The design of preemptible contracts is not only a question of profit maximization for the provider but also has significant implications for social welfare. By creating a market for reliability, these contracts can, in principle, lead to more efficient allocations of scarce computational resources, ensuring that capacity flows to its highest-value use, especially during periods of congestion. However, as is typical in markets characterized by monopoly power and information asymmetries, the private optimum achieved by a profit-maximizing firm is unlikely to coincide with the social optimum. Understanding the nature and magnitude of this inefficiency is the first step toward designing effective policy interventions.

In this section, we analyze the welfare properties of the kill-option menu. We first establish a utilitarian social planner's benchmark to identify the sources and magnitude of the inefficiency arising from monopoly screening. This decomposition reveals two distinct and intuitive distortions, one related to market segmentation (who is enrolled in the preemptible tier) and another related to the chosen level of reliability (how often interruptions occur). We then use this framework in the subsequent subsection to evaluate the efficacy of common policy instruments and to design a more sophisticated intervention that can, in principle, restore the first-best allocation.

5.1 Planner's Benchmark and Welfare Decomposition

To isolate the allocative inefficiencies of the monopoly outcome, we consider the problem faced by a benevolent social planner. This planner has the same technological capabilities as the monopolist and faces the same population of heterogeneous buyers. The planner's objective, however, is to maximize total utilitarian surplus, defined as the sum of all agents' payoffs. This includes the gross value buyers receive from the service, the operational value the provider gains from salvaged capacity, and the disutility buyers suffer from interruptions. Critically, transfers such as prices and refunds, which are central to the monopolist's problem, are mere redistributions of surplus from the planner's perspective and do not directly enter the welfare calculation.

The planner, like the monopolist, makes an allocative decision by choosing a cutoff $\varphi^\star$ and a hazard λ. For any such pair, total social welfare, W, is given by the sum of surplus generated in the

guaranteed and preemptible segments:

$$W(\varphi^\star, \lambda) \;=\; \underbrace{\int_{\varphi^\star}^{\bar{\varphi}} v f(\varphi) d\varphi}_{\text{Surplus from Guaranteed Users}} \;+\; \underbrace{\int_{\underline{\varphi}}^{\varphi^\star} \Big(v - \varphi\lambda + b(\lambda)\Big) f(\varphi) d\varphi}_{\text{Surplus from Preemptible Users}}. \tag{16}$$

This expression can be simplified. The gross value v is common to all buyers, so the first term is $(1 - F(\varphi^\star))v$. The second integral can be split, yielding $F(\varphi^\star)v - \lambda \int_{\underline{\varphi}}^{\varphi^\star} \varphi f(\varphi) d\varphi + F(\varphi^\star) b(\lambda)$. Combining these and factoring, we can express total welfare in a more compact and insightful form. Let $m(\varphi^\star) \equiv \mathbb{E}[\varphi \mid \varphi < \varphi^\star] = \frac{1}{F(\varphi^\star)} \int_{\underline{\varphi}}^{\varphi^\star} \varphi f(\varphi) d\varphi$ be the *average* interruption disutility of all buyers who are assigned to the preemptible class. The welfare function then becomes:

$$W(\varphi^\star, \lambda) \;=\; v \;+\; F(\varphi^\star)\Big(b(\lambda) - m(\varphi^\star)\,\lambda\Big). \tag{17}$$

Comparing this objective to the monopolist's profit function, $\Pi = v + F(\varphi^\star)(b(\lambda) - \varphi^\star\lambda)$, immediately reveals the fundamental conflict of interest. The planner is concerned with the *average* harm caused by interruptions across the entire preemptible pool, $m(\varphi^\star)$. The monopolist, in contrast, is disciplined by the harm to the *marginal* buyer, $\varphi^\star$, because it is this buyer's indifference condition that sets the price and determines the market segmentation. Since the marginal buyer is, by definition, the most sensitive type within the preemptible pool, we always have $\varphi^\star > m(\varphi^\star)$ for any non-degenerate interval. This wedge between the marginal and average type is the ultimate source of the allocative distortion.

The planner chooses $(\varphi^\star, \lambda)$ to maximize $W(\varphi^\star, \lambda)$. The first-order conditions for an interior solution characterize the socially optimal, or first-best, allocation.

Proposition 5.1 (The Planner's Solution)**.** *An interior first-best allocation* $(\varphi^{\star\,\mathrm{fb}}, \lambda^{\mathrm{fb}})$ *is characterized by the following two conditions:*

$$\textit{(Planner's Hazard Rule)} \qquad b'(\lambda^{\mathrm{fb}}) = m(\varphi^{\star\,\mathrm{fb}}) \tag{18}$$

$$\textit{(Planner's Cutoff Rule)} \qquad \frac{b(\lambda^{\mathrm{fb}})}{\lambda^{\mathrm{fb}}} = \varphi^{\star\,\mathrm{fb}} \tag{19}$$

Proof. To find the first-order conditions, we differentiate the welfare function in (17). The partial derivative with respect to λ is:

$$\frac{\partial W}{\partial \lambda} = F(\varphi^\star)\Big(b'(\lambda) - m(\varphi^\star)\Big).$$

Setting this to zero for an interior solution ($F(\varphi^\star) > 0$) yields the Planner's Hazard Rule. For the partial derivative with respect to $\varphi^\star$, we use the identity $d(F(\varphi)m(\varphi))/d\varphi = \varphi f(\varphi)$, which follows

from the definition of $m(\varphi)$. Differentiating (17) gives:

$$\frac{\partial W}{\partial \varphi^\star} = f(\varphi^\star)\Big(b(\lambda) - m(\varphi^\star)\lambda\Big) + F(\varphi^\star)\Big(-\lambda m'(\varphi^\star)\Big) = f(\varphi^\star)b(\lambda) - \lambda\frac{d}{d\varphi^\star}\Big(F(\varphi^\star)m(\varphi^\star)\Big).$$

Substituting the identity, we get $\frac{\partial W}{\partial \varphi^\star} = f(\varphi^\star)b(\lambda) - \lambda\varphi^\star f(\varphi^\star) = f(\varphi^\star)\Big(b(\lambda) - \varphi^\star\lambda\Big)$. Setting this to zero for an interior solution ($f(\varphi^\star) > 0$) and rearranging yields the Planner's Cutoff Rule. □

The economic logic of the planner's solution is clear and efficient. The Planner's Hazard Rule, $b'(\lambda) = m(\varphi^\star)$, equates the marginal benefit of an interruption (the marginal salvage value) to its marginal social cost. The marginal social cost is precisely the *average* disutility of the individuals at risk, as each is equally likely to be the one interrupted. The Planner's Cutoff Rule, $b(\lambda)/\lambda = \varphi^\star$, dictates the optimal size of the preemptible market. It expands the pool of preemptible users up to the point where the disutility of the marginal buyer, $\varphi^\star$, is exactly equal to the *average* salvage benefit generated per unit of hazard. Any buyer with a higher disutility would generate a net social loss if placed in the preemptible pool.

Decomposing the Monopoly Distortion. By comparing the planner's first-best conditions with the monopolist's optimality conditions from Proposition 3.1, we can precisely identify the nature and direction of the market failure. Recall the monopolist's rules:

$$\text{(Monopolist's Hazard Rule)} \quad b'(\lambda^{\mathrm{M}}) = \varphi^{\mathrm{M}}, \qquad \text{(Monopolist's Cutoff Rule)} \quad \frac{b(\lambda^{\mathrm{M}})}{\lambda^{\mathrm{M}}} - \varphi^{\mathrm{M}} = \frac{F(\varphi^{\mathrm{M}})}{f(\varphi^{\mathrm{M}})}.$$

This comparison immediately reveals two distinct wedges, which we can label as the enrollment distortion and the hazard distortion.

First, consider the *enrollment distortion*. The monopolist's cutoff rule includes the positive screening markup term $A(\varphi^M) \equiv F(\varphi^M)/f(\varphi^M) > 0$, which is absent from the planner's rule. This implies that for any given λ, the monopolist chooses a cutoff φ^M that is strictly lower than the one the planner would choose. This leads to over-enrollment: the monopolist pushes buyers into the preemptible class who, from a social standpoint, would generate more surplus in the guaranteed class. The monopolist does this because the marginal profit from selling to these types, even after the necessary discount, is positive, while the marginal social surplus is negative.

Next, we have the *hazard distortion*. The monopolist sets the hazard by equating marginal salvage to the disutility of the marginal type, $b'(\lambda^M) = \varphi^M$. The planner, however, would equate it to the average type, $b'(\lambda^{fb}) = m(\varphi^{fb})$. Since $\varphi^M > m(\varphi^M)$ and $b''(\lambda) < 0$, the monopolist's choice of hazard is inefficiently low (reliability is inefficiently high) *for the group it has chosen to serve*. To keep the price-sensitive marginal buyer φ^M in the preemptible class, the monopolist must promise a higher level of reliability than a planner, who is concerned with the lower average

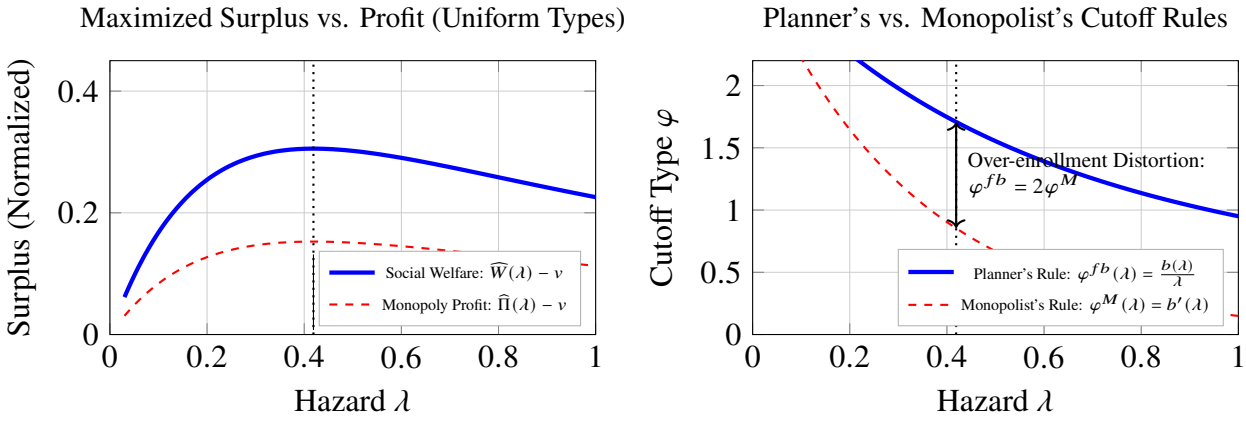
cost, would provide to that same group.

We can summarize the welfare loss at the monopoly optimum $(\varphi^{\mathrm{M}}, \lambda^{\mathrm{M}})$ by evaluating the gradient of the social welfare function at that point:

$$\left.\frac{\partial W}{\partial \varphi^{\star}}\right|_{\mathrm{M}} = f(\varphi^{\mathrm{M}})\Big(b(\lambda^{\mathrm{M}}) - \varphi^{\mathrm{M}}\lambda^{\mathrm{M}}\Big) = f(\varphi^{\mathrm{M}})\lambda^{\mathrm{M}}A(\varphi^{\mathrm{M}}) \;>\; 0, \tag{20}$$

$$\left.\frac{\partial W}{\partial \lambda}\right|_{\mathrm{M}} = F(\varphi^{\mathrm{M}})\Big(b'(\lambda^{\mathrm{M}}) - m(\varphi^{\mathrm{M}})\Big) = F(\varphi^{\mathrm{M}})\big(\varphi^{\mathrm{M}} - m(\varphi^{\mathrm{M}})\big) \;>\; 0. \tag{21}$$

Both gradients are strictly positive. This means that starting from the monopoly allocation, a social planner would wish to make two marginal changes: (1) increase the cutoff $\varphi^{\star}$ to shrink the inefficiently large preemptible market, and (2) increase the hazard λ to bring the reliability level closer to what is optimal for the (now smaller and less sensitive) group of preemptible users.



(a) For uniform types, concentrated welfare and profit functions are maximized at same hazard level.

(b) At the optimal hazard $\lambda^{\star}$, the monopolist's cutoff φ^{M} is strictly below the planner's φ^{fb}.

Figure 6: Decomposition of Monopoly Distortions under Uniform Types

A Special Case: Uniform Types. The interaction between the two distortions is particularly clear in the benchmark case of uniformly distributed types, where $m(\varphi) = \varphi/2$ and $A(\varphi) = \varphi$. The planner's first-order conditions become $b'(\lambda) = \varphi/2$ and $b(\lambda)/\lambda = \varphi$, which combine to give the same scalar identity as the monopolist's problem: $b(\lambda) = 2\lambda b'(\lambda)$.

Corollary 5.1 (Distortions under Uniform Types)**.** *If $\varphi \sim U[0, \bar{\varphi}]$ and an interior solution exists, then the monopolist and the planner choose the* ***same hazard level****, $\lambda^{\mathrm{M}} = \lambda^{\mathrm{fb}}$. The entire welfare loss comes from the enrollment distortion, with the monopolist's cutoff being exactly half of the planner's: $\varphi^{\mathrm{M}} = \frac{1}{2}\varphi^{\mathrm{fb}}$.*

This striking result shows that under the uniform distribution, the two distortions are perfectly offsetting in their effect on the hazard choice. The monopolist's tendency to under-provide hazard

for a given cutoff is exactly canceled out by its tendency to choose a cutoff that is too low. The inefficiency manifests entirely in the form of market segmentation: twice as many people are enrolled in the preemptible tier as is socially optimal. This provides a clean benchmark for thinking about policy: in markets that are approximately uniform, policy should focus on correcting the enrollment margin, not the reliability margin.

5.2 Policy Instruments and Second-Best Regulation

The welfare analysis of the previous section provides a clear diagnosis of the market failure: a profit-maximizing provider enrolls too many users in its preemptible tier and, for that group, offers an inefficiently high level of reliability. This diagnosis naturally leads to the question of remedies. What policy instruments might a regulator employ to mitigate these distortions and move the market outcome closer to the social optimum? In this section, we evaluate several potential interventions, ranging from sophisticated, information-intensive mechanisms to simpler, more practical rules of thumb.

Our analysis is grounded in the principle of second-best (Lipsey and Lancaster, 1956). We recognize that a regulator may face the same informational constraints as the monopolist (i.e., not observing buyer types directly) and may be limited to a restricted set of policy tools. We first describe an ideal policy, a transfer scheme based on sufficient statistics, that can, in theory, fully restore the first-best allocation. We then turn to a more pragmatic analysis of two commonly discussed interventions: reliability caps and price-display rules, deriving clear conditions under which these blunter instruments are likely to improve or worsen social welfare.

A Sufficient-Statistic Transfer that Attains the Planner's Optimum. The most direct way to correct a monopolist's behavior is to realign its incentives with those of the social planner. Our framework's reduction to the sufficient statistics of hazard (λ) and market share ($D = F(\varphi^\star)$) provides a powerful tool for doing so. A regulator who can observe these two market-level outcomes can, in principle, design a tax or subsidy scheme that induces the monopolist to choose the first-best allocation, without needing to regulate prices or refunds directly.

Consider a regulator who can levy a transfer $\mathcal{T}(\lambda, D)$ to the provider, where the amount of the transfer depends only on the observable hazard and the share of preemptible users. The provider's regulated profit function becomes:

$$\Pi^{\text{reg}}(\varphi^\star, \lambda) \;=\; \underbrace{F(\varphi^\star)\big(b(\lambda) - \varphi^\star\lambda\big)}_{\text{Monopoly Profit}} \;+\; \underbrace{\mathcal{T}\big(\lambda,\, F(\varphi^\star)\big)}_{\text{Corrective Transfer}} .$$

The goal is to design the function $\mathcal{T}(\cdot,\cdot)$ such that the provider, in maximizing this new objective, is led to choose the planner's optimal pair $(\lambda^{\text{fb}}, \varphi^{\text{fb}})$. This requires constructing a transfer whose marginal effects precisely offset the two monopoly distortions we identified.

Proposition 5.2 (Sufficient-Statistic Implementation of the Planner)**.** *There exists a transfer function $\mathcal{T}(\lambda, D)$, constructed solely from the observable market outcomes (λ, D) and knowledge of the type distribution $F(\cdot)$, that induces a profit-maximizing provider to choose the socially optimal allocation. A valid transfer is given by:*

$$\mathcal{T}(\lambda, D) \;\equiv\; D\,\sigma(\lambda, D) \;+\; \tau(\lambda, D), \tag{22}$$

where $\sigma(\lambda, D) = \lambda[\varphi(D) - m(\varphi(D))]$ and $\tau(\lambda, D) = \int_0^D [A(\varphi(u))\lambda - \sigma(\lambda, u)]du$, with $\varphi(D) = F^{-1}(D)$.

Proof. The firm's first-order conditions under the regulated profit function are $\frac{\partial \Pi^{\text{reg}}}{\partial \lambda} = 0$ and $\frac{\partial \Pi^{\text{reg}}}{\partial \varphi^\star} = 0$. The partial derivatives of the transfer $\mathcal{T}$ are constructed to exactly cancel the monopoly wedges in the firm's original first-order conditions. Specifically, $\frac{\partial \mathcal{T}}{\partial \lambda}$ is designed to offset the difference between $\varphi^\star$ and $m(\varphi^\star)$, while $\frac{\partial \mathcal{T}}{\partial D}$ is designed to eliminate the information rent term $A(\varphi^\star)$. A detailed derivation in the Appendix confirms that the resulting system of equations for the firm's choice variables is identical to the planner's first-order conditions (19). □

The economic intuition behind this mechanism is that the transfer provides two targeted corrections. The σ component acts as a per-user subsidy on the hazard, encouraging the provider to internalize the fact that the average social cost of interruption, $m(\varphi)$, is lower than the private marginal cost, φ. The τ component acts as an enrollment tax, discouraging the provider from over-expanding the preemptible market in its quest to extract rents. The power of this approach is its informational efficiency: the regulator does not need to know the provider's proprietary salvage function $b(\lambda)$. It only needs to estimate the demand-side primitive $F(\cdot)$ and then commit to a transfer schedule based on observable market outcomes.

Simple Rules: Reliability Caps. While the sufficient-statistic transfer is theoretically elegant, implementing it may be complex. Regulators often prefer simpler, more direct interventions. A common example is a reliability cap, which we model as an upper bound $\lambda \le \bar{\lambda}$. As discussed in Section 3.4, a binding cap forces the provider to set $\lambda = \bar{\lambda}$ and adjust its cutoff $\varphi^\star$ in response. The monopolist's best-response cutoff, $\varphi^\star(\lambda)$, is a decreasing function of λ.

The welfare effect of tightening such a cap is therefore ambiguous. A tighter cap (lower $\bar{\lambda}$) has two opposing effects on welfare:

1. **Direct Hazard Effect (Negative):** It forces the provider to reduce λ, moving it further away from the socially optimal level and exacerbating the "under-interruption" distortion.

2. **Indirect Cutoff Effect (Positive):** By making the preemptible contract more attractive, it forces the provider to raise its cutoff $\varphi^\star(\lambda)$, moving it closer to the social optimum and mitigating the "over-enrollment" distortion.

The net effect on welfare depends on which of these two forces is stronger at the margin. We can derive a precise condition for when a cap is locally welfare-improving.

Proposition 5.3 (When Do Caps Raise Welfare at the Margin?)**.** *Starting from the unconstrained monopoly allocation* $(\varphi^{\mathrm{M}}, \lambda^{\mathrm{M}})$*, a marginal tightening of a reliability cap (a small decrease in* λ*) increases social welfare if and only if:*

$$\underbrace{\varphi^{\mathrm{M}} - m(\varphi^{\mathrm{M}})}_{\textit{Interruption Wedge}} \; < \; \underbrace{\frac{A(\varphi^{\mathrm{M}})}{1 + A'(\varphi^{\mathrm{M}})}}_{\textit{Screening Wedge}} . \tag{23}$$

Proof. The proof involves evaluating the total derivative of the welfare function, $\frac{d}{d\lambda} W(\varphi^{\star}(\lambda), \lambda)$, at the monopoly solution. This derivative captures both the direct effect through λ and the indirect effect through the monopolist's adjustment of $\varphi^{\star}(\lambda)$. The sign of this derivative, and thus the desirability of tightening the cap, hinges on the comparison in (23). A full derivation is in the Appendix. □

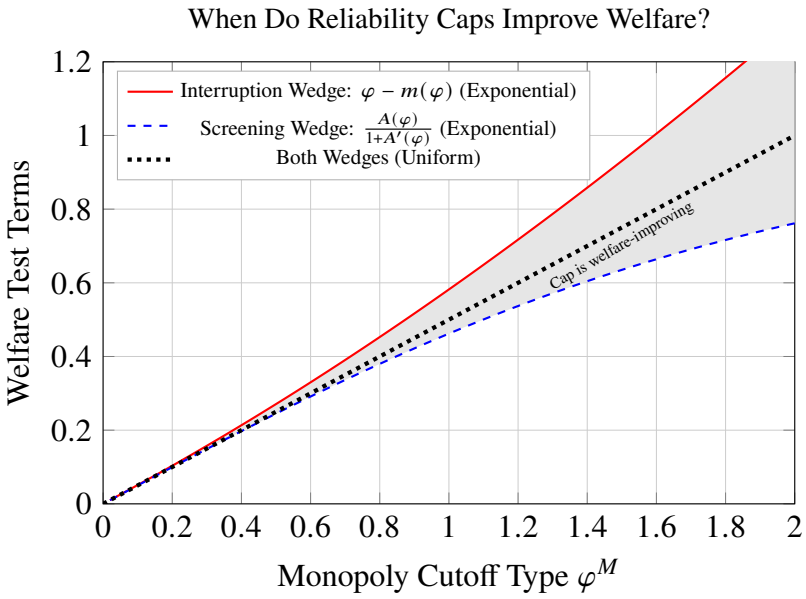


Figure 7: The geometric interpretation of the cap criterion from Proposition 5.3. For uniform types, the two wedges that drive the welfare trade-off are identical (dotted line), making caps locally neutral. For other distributions like the exponential, caps are locally welfare-improving if the monopolist's cutoff φ^M falls in the shaded region where the screening wedge exceeds the interruption wedge.

This condition provides a clear, testable criterion for the desirability of reliability caps. The "Interruption Wedge" on the left captures the magnitude of the hazard distortion, while the "Screening Wedge" on the right (related to the curvature of the information rent function) captures the

magnitude of the enrollment distortion. A cap is beneficial if the harm from over-enrollment is sufficiently large relative to the harm from under-interruption. This is more likely to be the case for type distributions with a rapidly increasing inverse hazard rate (a large A'), such as those with heavy right tails. This highlights the perils of simple, blunt instruments in a second-best world. The condition in Proposition 5.3 provides a clear, testable criterion for the desirability of reliability caps, which we visualize in Figure 7. The figure plots the two competing wedges against each other for different underlying type distributions. For the benchmark case of a uniform distribution, the two sides of the inequality are exactly equal, meaning reliability caps are first-order neutral at the monopoly optimum, a result consistent with Corollary 5.1. For a distribution with a heavier tail like the exponential, however, the screening wedge can dominate the interruption wedge, creating a region where a marginal tightening of the cap would be welfare-improving. This demonstrates that the appropriateness of such a policy is not a universal constant but an empirical question that depends on the specific shape of market demand.

Simple Rules: Display Constraints. In contrast to the ambiguity of reliability caps, the welfare effects of tightening price-display rules are unambiguous. As shown in Proposition 3.8, when constraints on posted prices or discounts bind, they induce the monopolist to choose both a lower hazard λ and a lower cutoff $\varphi^\star$.

Proposition 5.4 (Display Bounds are Locally Welfare-Worsening)**.** *Starting from the unconstrained monopoly allocation, a marginal tightening of a price-display rule is unambiguously welfare-reducing.*

The proof follows directly from the welfare gradients derived in (20) and (21). At the monopoly point, social welfare is strictly increasing in both λ and $\varphi^\star$. Since a tighter display rule forces the monopolist to decrease both variables, it necessarily moves the allocation further away from the social optimum, reducing total surplus. The justification for such policies, if any, must therefore lie outside the utilitarian framework, perhaps in considerations of fairness, transparency, or consumer psychology that are not captured in our model.

5.3 Congestion, Externalities, and Welfare with Spillovers

Our welfare analysis thus far has maintained the assumption that the only social costs of interruption are the private disutilities, φ, borne by the affected buyers. In a complex, shared computing environment, this may not be the case. High rates of interruption could generate negative externalities that affect other users or the overall system performance. For instance, the simultaneous termination and checkpointing of thousands of preemptible jobs could cause a surge in network traffic or load on shared storage systems, creating "control plane congestion" that might even degrade the performance of guaranteed instances.

We can incorporate such spillovers into our welfare framework by adding a reduced-form externality cost, $\Xi(\lambda, D)$, to the social welfare function, where $D = F(\varphi^\star)$ is the share of preemptible users. We assume this cost is increasing in both the hazard rate and the number of users at risk ($\Xi_\lambda > 0, \Xi_D > 0$). The planner's objective becomes:

$$W^{\text{ext}}(\varphi^\star, \lambda) \;=\; v \;+\; F(\varphi^\star)\Big(b(\lambda) - m(\varphi^\star)\,\lambda\Big) \;-\; \Xi\big(\lambda,\, F(\varphi^\star)\big).$$

The presence of this externality modifies the planner's first-best conditions. The socially optimal hazard and cutoff will now be lower than in the baseline case to account for these additional system-wide costs. The new first-order conditions become:

$$\boxed{b'(\lambda^{\text{fb}}) = m(\varphi^{\text{fb}}) + \frac{\Xi_\lambda}{F(\varphi^{\text{fb}})}} \qquad \text{and} \qquad \boxed{\frac{b(\lambda^{\text{fb}})}{\lambda^{\text{fb}}} = \varphi^{\text{fb}} + \frac{\Xi_D}{\lambda^{\text{fb}}}}. \tag{24}$$

The externality effectively increases the marginal social cost of both the hazard and enrollment, pushing the planner's target allocation towards a lower-risk, lower-enrollment regime.

Interestingly, the presence of negative externalities can reduce the gap between the monopoly and the social optimum. The monopolist, driven by its own private incentives, already chooses a hazard rate that is inefficiently low from a baseline welfare perspective. The externality pushes the social optimum in the same direction, meaning the monopolist's choice may be "accidentally" closer to the socially desirable level. This suggests that in systems with significant interruption externalities, the welfare loss from monopoly may be less severe than our baseline analysis suggests, and the policy case for interventions like reliability caps (which further reduce the hazard) becomes stronger. The sufficient-statistics transfer can also be augmented with a simple Pigouvian tax component to fully internalize these external costs.

6 Conclusion

This paper reframes preemptible cloud instances as posted contracts that embed a provider "kill option," with two design levers at center stage: an interruption hazard chosen by the operator and a per-event refund paid to buyers. The analysis shows that refunds are implementational rather than value-creating; once prices adjust to preserve the selection margin, profits depend only on the hazard and on the cutoff type that separates preemptible from guaranteed users. This reduction delivers clean characterizations of optimal menus, a tight equivalence to damaged-goods quality design, and a mapping to simple priority auctions that implement the same allocation and expected payments.

By isolating the hazard and the cutoff as sufficient statistics, the model yields transparent managerial guidance. When the shadow value of capacity rises, the optimal policy increases

the hazard and, through the induced cutoff, shifts more interruption-tolerant workloads into the preemptible tier. Generous refunds help implement these movements without altering profitability, unless display rules or refund floors restrict how prices and rebates can move. In such regulated settings, the firm should anticipate lower hazards and narrower posted gaps, and recognize that those rules blunt the very screening forces that make preemptible offerings attractive.

The welfare decomposition clarifies where market power bites. Relative to a planner who internalizes the average disutility among preemptible users and the marginal value of reclaimed capacity, the monopolist enrolls too many buyers into the risky tier and then sets reliability too high for that enrollment. These two wedges organize policy. Reliability caps may help when type distributions are heavy-tailed and the planner's cutoff is far above the monopolist's; they are locally neutral under distributions that mimic a uniform benchmark. Display constraints that cap posted discounts or mandate high refunds are generically welfare-worsening at the margin because they push both margins in the wrong direction. A simple sufficient-statistics transfer, based only on the observed hazard and the observed share of buyers in the preemptible tier, decentralizes the planner's allocation without micromanaging prices or the refund schedule.

The framework is also diagnostic for empirical work. It furnishes testable implications relating hazards, refunds, and enrollment shares to observable congestion proxies such as utilization, queue backlogs, or incident headroom. It suggests identification strategies that use posted discount experiments, reliability sweeps, and exogenous shifts in capacity value to trace out the sufficient statistics. And it points to measurement of salvage via internal shadow prices from capacity planning or autoscaling systems, which can be cross-validated against observed kill rates and realized refunds.

Several extensions are natural. On the demand side, allowing correlation between baseline value and interruption disutility invites multidimensional screening and raises questions about when a two-option menu remains near-optimal. On the risk dimension, buyers' risk aversion can be handled by de-randomizing refunds into deterministic credits without changing the core design. On the supply side, multi-tier hazard ladders, duration commitments, and renewal options would connect the kill-option logic to longer-term contracts. Competition across regions and providers would bring platform considerations to the fore, including fairness constraints, carbon-aware scheduling, and environmental externalities. Finally, richer dynamics, learning about workloads' tolerance, forecasting bursts, or coupling hazards to carbon intensity, can be folded into the salvage microfoundation without disturbing the core sufficient-statistics structure.

Taken together, the kill-option view unifies versioning and priority under operational congestion. It explains why preemptible offerings exist, how to tune them as conditions change, what policies improve or degrade welfare, and how to take the theory to data using objects that platforms already record. We hope the simplicity of the sufficient-statistics approach makes the results useful for both

researchers and practitioners who design, regulate, and rely on cloud reliability menus at scale.

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Appendix A: Proofs and Additional Material for Section 2

This appendix provides the formal proofs and supplementary technical details for the results presented in Section 2 of the main text. We adopt the notation and assumptions introduced therein.

A.1. Buyer Self-Selection: Proof of the Cutoff Rule

The main text establishes that buyer choice is governed by a simple cutoff rule. Here, we formalize this result.

Proposition .1 (Cutoff Selection)**.** *Fix a menu* $(p_G, 0)$ *and* (p_P, λ, r) *with* $\lambda > 0$*. There exists a unique cutoff* $\varphi^\star$ *given by* $\varphi^\star = \frac{p_G - p_P}{\lambda} + r$ *such that buyers with* $\varphi \geq \varphi^\star$ *choose the guaranteed contract and those with* $\varphi < \varphi^\star$ *choose the preemptible contract.*

Proof. A buyer of type φ chooses the guaranteed contract (G) over the preemptible contract (P) if $U_G(\varphi) \geq U_P(\varphi)$. The utility difference is:

$$\Delta(\varphi) \equiv U_G(\varphi) - U_P(\varphi) = (v - p_G) - \big(v - p_P - \varphi\lambda + r\lambda\big) = (p_P - p_G) + (\varphi - r)\lambda.$$

(i) If $\lambda > 0$, then $\Delta(\varphi)$ is a strictly increasing and continuous function of φ, with slope λ. Since its range is $(-\infty, \infty)$, there must exist a unique type $\varphi^\star$ for which $\Delta(\varphi^\star) = 0$. Solving for this type yields:

$$(p_P - p_G) + (\varphi^\star - r)\lambda = 0 \quad \Longrightarrow \quad \varphi^\star = \frac{p_G - p_P}{\lambda} + r.$$

By the monotonicity of $\Delta(\varphi)$, for any $\varphi > \varphi^\star$, we have $\Delta(\varphi) > 0$, implying $U_G(\varphi) > U_P(\varphi)$, so the buyer chooses G. For any $\varphi < \varphi^\star$, we have $\Delta(\varphi) < 0$, implying $U_G(\varphi) < U_P(\varphi)$, so the buyer chooses P. Because the type distribution F is continuous, the set of indifferent types at $\varphi^\star$ has measure zero, and the stated cutoff rule holds almost everywhere.

(ii) If $\lambda = 0$, the contracts differ only in price. $\Delta(\varphi) = p_P - p_G$, which is constant. Buyers will choose G if $p_G < p_P$, P if $p_P < p_G$, and will be indifferent if prices are equal. This completes the proof. □

A.2. The Seller's Problem: Proof of Program Reduction (Proposition 2.1)

Proposition 2.1 in the main text is the central simplification of our model. It states that the seller's four-dimensional choice of menu parameters can be reduced to a two-dimensional choice of allocative variables. We provide a detailed proof with all algebraic steps made explicit.

Proof of Proposition 2.1. The seller chooses (p_G, p_P, λ, r) to maximize total expected profit. Given a cutoff $\varphi^\star$ induced by the menu, the demands are $D_P = F(\varphi^\star)$ for the preemptible contract and $D_G = 1 - F(\varphi^\star)$ for the guaranteed contract. Total profit is the sum of profits from

each segment:

$$\begin{aligned}\Pi &= D_G \cdot \pi_G + D_P \cdot \pi_P \\ &= \big(1 - F(\varphi^\star)\big) \cdot p_G + F(\varphi^\star) \cdot \big(p_P + b(\lambda) - r\lambda\big)\end{aligned} \tag{25}$$

From the buyer's indifference condition (Equation (4) in the main text), the marginal type $\varphi^\star$ satisfies:

$$U_G(\varphi^\star) = U_P(\varphi^\star)$$

Substituting the utility expressions:

$$v - p_G = v - p_P - \varphi^\star\lambda + r\lambda$$

Simplifying:

$$-p_G = -p_P - \varphi^\star\lambda + r\lambda$$

$$p_G - p_P = \varphi^\star\lambda - r\lambda = \lambda(\varphi^\star - r)$$

Therefore, we can express the preemptible price as:

$$p_P = p_G - \lambda(\varphi^\star - r) \tag{26}$$

Substituting equation (26) into the profit expression (25):

$$\begin{aligned}\Pi &= \big(1 - F(\varphi^\star)\big)p_G + F(\varphi^\star)\Big[\big(p_G - \lambda(\varphi^\star - r)\big) + b(\lambda) - r\lambda\Big] \\ &= \big(1 - F(\varphi^\star)\big)p_G + F(\varphi^\star)\big[p_G - \lambda\varphi^\star + r\lambda + b(\lambda) - r\lambda\big] \\ &= \big(1 - F(\varphi^\star)\big)p_G + F(\varphi^\star)\big[p_G - \lambda\varphi^\star + b(\lambda)\big]\end{aligned}$$

Note that the refund terms $r\lambda$ cancel out in the preemptible profit expression. Expanding the expression:

$$\begin{aligned}\Pi &= p_G - F(\varphi^\star)p_G + F(\varphi^\star)p_G - F(\varphi^\star)\lambda\varphi^\star + F(\varphi^\star)b(\lambda) \\ &= p_G + F(\varphi^\star)\big[p_G - p_G - \lambda\varphi^\star + b(\lambda)\big] \\ &= p_G + F(\varphi^\star)\big[b(\lambda) - \varphi^\star\lambda\big]\end{aligned}$$

Therefore, the reduced-form profit function is:

$$\Pi = p_G + F(\varphi^\star)\big[b(\lambda) - \varphi^\star\lambda\big] \tag{27}$$

The guaranteed price p_G enters the profit function additively and independently of $(\lambda, \varphi^\star)$. To maximize profit, the seller should set p_G as high as possible, subject to the participation constraint of buyers who choose the guaranteed contract. For a type $\varphi \geq \varphi^\star$ who chooses the guaranteed contract:

$$U_G(\varphi) = v - p_G$$

The participation (individual rationality) constraint requires:

$$U_G(\varphi) \geq 0 \implies v - p_G \geq 0 \implies p_G \leq v$$

Since profit is increasing in p_G and the constraint binds at $p_G = v$, the optimal guaranteed price is:

$$p_G^\star = v$$

Substituting $p_G^\star = v$ into equation (27):

$$\Pi = v + F(\varphi^\star)\big[b(\lambda) - \varphi^\star \lambda\big]$$

Since v is a constant, maximizing total profit is equivalent to maximizing the second term. The seller's problem therefore reduces to:

$$\max_{\lambda \in [0,1],\ \varphi^\star \in [\underline{\varphi}, \bar{\varphi}]} F(\varphi^\star)\big[b(\lambda) - \varphi^\star \lambda\big]$$

This is a two-dimensional optimization problem in the sufficient statistics $(\lambda, \varphi^\star)$. For any solution $(\lambda^\star, \varphi^\star)$ to the reduced program, we can implement it by setting:

- $p_G^\star = v$
- Any pair (p_P, r) satisfying $p_P = v - (\varphi^\star - r)\lambda^\star$

We can see that refund r does not appear in the reduced profit function. It serves only to implement a desired cutoff $\varphi^\star$ for a given hazard λ. Any value of r that keeps prices non-negative and satisfies other practical constraints can be chosen without affecting the allocation or profit. This completes the proof that the four-dimensional menu design problem reduces to a two-dimensional optimization over $(\lambda, \varphi^\star)$. □

A.3. Microfoundation for Salvage Value: Proof of Properties

Section 2.3 of the main text provides a microfoundation for the salvage function $b(\lambda)$ based on a state-contingent targeting problem. Here we provide the formal proof for the properties of $b(\lambda)$ derived from that problem.

Proposition .2 (Optimal Targeting and Properties of $b(\lambda)$)**.** *Let the shadow value of capacity $\mu(s)$ be integrable over the state space $(\mathcal{S}, m)$. The salvage function $b(\lambda)$ derived from the targeting problem in Equation (6) has the following properties:*

1. *The optimal hazard schedule $h^\star(s)$ is a threshold rule: there exists a cutoff $\kappa(\lambda)$ such that $h^\star(s) = 1$ if $\mu(s) > \kappa(\lambda)$ and $h^\star(s) = 0$ if $\mu(s) < \kappa(\lambda)$.*

2. *The function $b(\lambda)$ is increasing and concave on $[0, 1]$.*

3. *If the distribution of μ is continuous, $b(\lambda)$ is differentiable and its derivative is the threshold: $b'(\lambda) = \kappa(\lambda)$.*

Proof. The problem is to choose a measurable function $h : \mathcal{S} \to [0, 1]$ to solve:

$$\max_{h(\cdot)} \int_{\mathcal{S}} \mu(s)h(s)m(ds) \quad \text{subject to} \quad \int_{\mathcal{S}} h(s)m(ds) = \lambda.$$

This is a standard isoperimetric problem in the calculus of variations, or a continuous knapsack problem.

(1) **Threshold Structure:** We can form the Lagrangian with multiplier κ:

$$\mathcal{L}(h, \kappa) = \int_{\mathcal{S}} \mu(s)h(s)m(ds) - \kappa\left(\int_{\mathcal{S}} h(s)m(ds) - \lambda\right) = \int_{\mathcal{S}} \Big(\mu(s) - \kappa\Big)h(s)m(ds) + \kappa\lambda.$$

To maximize this integral pointwise subject to $h(s) \in [0, 1]$, the optimal choice $h^\star(s)$ must follow a "on-off" solution. For each state s, the term $(\mu(s) - \kappa)h(s)$ is maximized by setting $h^\star(s) = 1$ if $\mu(s) - \kappa > 0$ and $h^\star(s) = 0$ if $\mu(s) - \kappa < 0$. If $\mu(s) = \kappa$, any $h(s) \in [0, 1]$ is optimal. The multiplier κ is chosen to ensure the budget constraint $\int h^\star(s)m(ds) = \lambda$ is met. This defines κ as a function of λ, $\kappa(\lambda)$, and establishes the threshold structure.

(2) **Monotonicity and Concavity:** Let $\mu^\downarrow(q)$ be the decreasing rearrangement of $\mu(s)$, which is the quantile function of μ evaluated at $1 - q$. The threshold policy implies that for a given λ, the value of the integral is the sum of the top λ fraction of μ values. This can be written as:

$$b(\lambda) = \int_0^\lambda \mu^\downarrow(q)dq.$$

Since $\mu(s) \geq 0$, $\mu^\downarrow(q) \geq 0$, which implies $b'(\lambda) = \mu^\downarrow(\lambda) \geq 0$, so $b(\lambda)$ is increasing. Furthermore, since $\mu^\downarrow(q)$ is a non-increasing function of q, its integral, $b(\lambda)$, must be a concave function of its upper limit λ.

(3) **Derivative:** By the Fundamental Theorem of Calculus (for Lebesgue integrals), if $b(\lambda) = \int_0^\lambda \mu^\downarrow(q)dq$, then $b'(\lambda) = \mu^\downarrow(\lambda)$ almost everywhere. If the distribution of μ is continuous, then its

quantile function is continuous, meaning $\mu^{\downarrow}$ is continuous, and the derivative exists everywhere. The threshold $\kappa(\lambda)$ is defined by the condition $m(\{s : \mu(s) \geq \kappa(\lambda)\}) = \lambda$. This means $\kappa(\lambda)$ is precisely the $(1-\lambda)$-quantile of the distribution of μ, which is the definition of $\mu^{\downarrow}(\lambda)$. Thus, $b'(\lambda) = \kappa(\lambda)$. □

Appendix B: Proofs and Additional Material for Section 3

This appendix provides the formal proofs and supplementary technical results that underpin the characterization of the optimal menu in Section 3. We maintain the primitives and notation from the main text, in particular the reduced profit function:

$$\Phi(\varphi^\star, \lambda) = F(\varphi^\star)\big(b(\lambda) - \varphi^\star\lambda\big),$$

where F is the CDF of types and b is the salvage function.

B.1. Proofs for Interior and Boundary Characterizations

This section contains the detailed proofs for the propositions and lemmas in Sections 3.1 and 3.2 of the main text, covering the interior solution, extreme hazard conditions, and behavior at the boundaries of the choice space.

Proof of Proposition 3.1 (Interior First-Order Characterization). The seller's objective is to maximize $\Phi(\varphi^\star, \lambda)$ over $(\varphi^\star, \lambda) \in [\underline{\varphi}, \bar{\varphi}] \times [0, 1]$. For an interior solution $(\varphi^\star, \lambda) \in (\underline{\varphi}, \bar{\varphi}) \times (0, 1)$, the first-order necessary conditions are $\frac{\partial \Phi}{\partial \lambda} = 0$ and $\frac{\partial \Phi}{\partial \varphi^\star} = 0$.

The partial derivative with respect to λ is:

$$\frac{\partial \Phi}{\partial \lambda} = F(\varphi^\star)\big(b'(\lambda) - \varphi^\star\big).$$

Since $F(\varphi^\star) > 0$ for an interior cutoff, setting the derivative to zero yields:

$$b'(\lambda) - \varphi^\star = 0 \implies \varphi^\star = b'(\lambda).$$

The partial derivative with respect to $\varphi^\star$ is:

$$\frac{\partial \Phi}{\partial \varphi^\star} = f(\varphi^\star)\big(b(\lambda) - \varphi^\star\lambda\big) - F(\varphi^\star)\lambda.$$

Setting the derivative to zero yields:

$$f(\varphi^\star)\big(b(\lambda) - \varphi^\star\lambda\big) = F(\varphi^\star)\lambda.$$

Since $f(\varphi^\star) > 0$ and $\lambda > 0$ at an interior solution, we can rearrange to get:

$$\frac{b(\lambda)}{\lambda} - \varphi^\star = \frac{F(\varphi^\star)}{f(\varphi^\star)}.$$

This establishes the two conditions in the proposition. For the converse, if a pair $(\varphi^\star, \lambda)$ solves these equations and the Hessian matrix of Φ is negative semidefinite at that point, it constitutes a local maximum. □

Proof of Proposition 3.2 (Extreme Hazards). The provider's incentive to change the hazard is determined by the sign of $\partial\Phi/\partial\lambda = F(\varphi^\star)\big(b'(\lambda) - \varphi^\star\big)$.

(1) **No Preemption** ($\lambda^\star = 0$): We are given that $c_0 \equiv b'(0^+) \leq \underline{\varphi}$. By the concavity of $b(\cdot)$, we know that $b'(\lambda)$ is a non-increasing function of λ. Therefore, for any $\lambda \in [0, 1]$, we have $b'(\lambda) \leq b'(0^+) = c_0$. For any feasible cutoff $\varphi^\star \in [\underline{\varphi}, \bar{\varphi}]$, it must be that $\varphi^\star \geq \underline{\varphi}$. Combining these, we have $b'(\lambda) \leq c_0 \leq \underline{\varphi} \leq \varphi^\star$. This implies that the term $b'(\lambda) - \varphi^\star$ is non-positive for all feasible $(\varphi^\star, \lambda)$. Since $F(\varphi^\star) \geq 0$, the derivative $\partial\Phi/\partial\lambda$ is non-positive everywhere. Thus, the profit function $\Phi(\varphi^\star, \lambda)$ is weakly decreasing in λ for any fixed $\varphi^\star$, and the optimum must occur at the lowest possible value, $\lambda^\star = 0$. At this point, profit is $\Phi(\varphi^\star, 0) = F(\varphi^\star)(b(0) - 0) = 0$.

(2) **Maximal Preemption** ($\lambda^\star = 1$): We are given that $c_1 \equiv b'(1^-) \geq \bar{\varphi}$. By concavity, $b'(\lambda) \geq b'(1^-) = c_1$ for any $\lambda \in [0, 1]$. For any feasible cutoff $\varphi^\star \in [\underline{\varphi}, \bar{\varphi}]$, it must be that $\varphi^\star \leq \bar{\varphi}$. Combining these gives $b'(\lambda) \geq c_1 \geq \bar{\varphi} \geq \varphi^\star$. This implies that $b'(\lambda) - \varphi^\star$ is non-negative for all feasible $(\varphi^\star, \lambda)$. The derivative $\partial\Phi/\partial\lambda$ is therefore non-negative everywhere, meaning profit is weakly increasing in λ. The optimum must occur at the highest possible value, $\lambda^\star = 1$. □

Proof of Proposition 3.3 (Linear Salvage). When $b(\lambda) = \beta\lambda$, the profit function becomes $\Phi(\varphi^\star, \lambda) = F(\varphi^\star)\big(\beta\lambda - \varphi^\star\lambda\big) = \lambda F(\varphi^\star)(\beta - \varphi^\star)$. For any given cutoff $\varphi^\star$, this function is linear in λ. The provider's choice of λ is therefore binary:

- If the term $F(\varphi^\star)(\beta - \varphi^\star)$ is positive, the provider sets $\lambda = 1$.
- If the term is negative, the provider sets $\lambda = 0$.
- If the term is zero, the provider is indifferent.

If $\beta \leq \underline{\varphi}$, then for any feasible $\varphi^\star \in [\underline{\varphi}, \bar{\varphi}]$, we have $\beta - \varphi^\star \leq 0$. Since $F(\varphi^\star) \geq 0$, the entire term is non-positive, so the optimal choice is $\lambda^\star = 0$. If $\beta > \underline{\varphi}$, the provider can choose a cutoff $\varphi^\star$ such that $\underline{\varphi} \leq \varphi^\star < \beta$. For such a choice, $F(\varphi^\star) > 0$ (unless $\underline{\varphi}$ is the only type, which we rule out) and $\beta - \varphi^\star > 0$. This makes the profit function strictly increasing in λ, so the provider must set $\lambda^\star = 1$. Conditional on $\lambda = 1$, the provider then chooses $\varphi^\star$ to solve $\max_{\varphi \in [\underline{\varphi}, \bar{\varphi}]} F(\varphi)(\beta - \varphi)$. □

B.2. Proofs for Canonical Families

This section provides the detailed derivations for the solutions under uniform types presented in Section 3.3.

Proof of Proposition 3.4 (Uniform Types, Linear Salvage). From Proposition **??**, we know that if $\beta > 0$, the optimal hazard is $\lambda^\star = 1$. The provider's problem reduces to choosing $\varphi^\star$ to maximize $g(\varphi) = F(\varphi)(\beta - \varphi)$. With $\varphi \sim U[0, \bar{\varphi}]$, this is $g(\varphi) = (\varphi/\bar{\varphi})(\beta - \varphi)$ for $\varphi \in [0, \bar{\varphi}]$. This is a simple concave quadratic in φ. The first-order condition is $g'(\varphi) = (1/\bar{\varphi})(\beta - 2\varphi) = 0$, which yields an unconstrained maximizer of $\varphi = \beta/2$. The optimal choice must lie within the feasible interval $[0, \bar{\varphi}]$, so we must truncate the solution:

$$\varphi^\star = \min\left\{\frac{\beta}{2}, \bar{\varphi}\right\}.$$

We then calculate the resulting profit, $\Pi - v = \Phi(\varphi^\star, 1) = (\varphi^\star/\bar{\varphi})(\beta - \varphi^\star)$:

- If $\beta/2 \le \bar{\varphi}$ (i.e., $\beta \le 2\bar{\varphi}$), then $\varphi^\star = \beta/2$. Profit is $(\frac{\beta/2}{\bar{\varphi}})(\beta - \frac{\beta}{2}) = \frac{\beta^2}{4\bar{\varphi}}$.
- If $\beta/2 > \bar{\varphi}$ (i.e., $\beta > 2\bar{\varphi}$), then the boundary solution $\varphi^\star = \bar{\varphi}$ is optimal. Profit is $(\frac{\bar{\varphi}}{\bar{\varphi}})(\beta - \bar{\varphi}) = \beta - \bar{\varphi}$.

This confirms the expressions in the proposition. □

Proof of Proposition 3.5 (Uniform Types, Exponential Salvage). For an interior solution with uniform types, the condition is $b(\lambda) = 2\lambda b'(\lambda)$. With $b(\lambda) = \beta(1 - e^{-a\lambda})$ and $b'(\lambda) = \beta a e^{-a\lambda}$, this becomes:

$$\beta(1 - e^{-a\lambda}) = 2\lambda(\beta a e^{-a\lambda}) \implies 1 - e^{-a\lambda} = 2a\lambda e^{-a\lambda} \implies e^{a\lambda} = 1 + 2a\lambda.$$

Let $x = a\lambda$. The equation is $e^x = 1 + 2x$. We can rewrite this as $1 = (1 + 2x)e^{-x}$. Let $y = 1 + 2x$, so $x = (y - 1)/2$. Substituting gives $1 = ye^{-(y-1)/2} = ye^{1/2}e^{-y/2}$. Rearranging yields $-\frac{1}{2}e^{-1/2} = -\frac{y}{2}e^{-y/2}$. This is of the form $z = we^w$, whose solution is given by the Lambert W function, $w = W(z)$. Here, $w = -y/2$ and $z = -e^{-1/2}/2$. The relevant branch is W_{-1} for a positive solution x. So, $-y/2 = W_{-1}(-e^{-1/2}/2)$. Substituting back for x:

$$x^\star = \frac{y - 1}{2} = -W_{-1}\left(-\frac{e^{-1/2}}{2}\right) - \frac{1}{2} \approx 1.25643.$$

This gives the candidate interior solution $\lambda^\circ = x^\star/a$. The corresponding cutoff is $\varphi^{\star\circ} = b'(\lambda^\circ) = \beta a e^{-a\lambda^\circ} = \beta a e^{-x^\star} = \frac{\beta a}{1+2x^\star}$. This is the interior optimum if it is feasible, i.e., $\lambda^\circ \le 1 \implies a \ge x^\star$ and $\varphi^{\star\circ} \le \bar{\varphi}$.

The profit at the interior solution is $\Pi - v = \Phi = (\varphi^\star/\bar{\varphi})(b(\lambda) - \varphi^\star\lambda)$. Using the interior conditions $b(\lambda) = 2\lambda\varphi^\star$, this simplifies to $\Pi - v = (\varphi^\star/\bar{\varphi})(\lambda\varphi^\star) = \lambda(\varphi^\star)^2/\bar{\varphi}$. Substituting the solutions for $\lambda^\star$ and $\varphi^\star$:

$$\Pi - v = \frac{1}{\bar{\varphi}}\left(\frac{x^\star}{a}\right)\left(\frac{\beta a}{1+2x^\star}\right)^2 = \frac{x^\star\beta^2 a}{(1+2x^\star)^2\bar{\varphi}}.$$

If the candidate is not feasible, the optimum must lie on a boundary, as characterized in §3.2. For example, if $a < x^\star$, then $\lambda^\circ > 1$. Since Φ is concave in λ for a fixed $\varphi^\star$, the optimum must be at $\lambda = 1$. The logic for the other boundary cases follows from applying the general lemmas from §3.2. □

Proof of Proposition 3.6 (Power Salvage). With $b(\lambda) = \beta\lambda^\gamma$, the interior condition $b(\lambda) = 2\lambda b'(\lambda)$ becomes:

$$\beta\lambda^\gamma = 2\lambda(\beta\gamma\lambda^{\gamma-1}) \implies \beta\lambda^\gamma = 2\beta\gamma\lambda^\gamma \implies 1 = 2\gamma.$$

(1) If $\gamma = 1/2$, this condition is satisfied for all $\lambda > 0$. This means that for any λ, if we set the cutoff according to the first FOC, $\varphi^\star = b'(\lambda) = \beta/(2\sqrt{\lambda})$, the second FOC is automatically satisfied. The profit for any such pair is:

$$\Phi(\varphi^\star, \lambda) = \frac{\varphi^\star}{\bar{\varphi}}(b(\lambda) - \varphi^\star\lambda) = \frac{\beta/(2\sqrt{\lambda})}{\bar{\varphi}}\left(\beta\sqrt{\lambda} - \frac{\beta}{2\sqrt{\lambda}}\lambda\right) = \frac{\beta}{2\bar{\varphi}\sqrt{\lambda}}\left(\frac{\beta\sqrt{\lambda}}{2}\right) = \frac{\beta^2}{4\bar{\varphi}}.$$

This profit is constant, independent of λ. The provider is indifferent over any pair on this locus, subject to the feasibility constraints $\lambda \in (0, 1]$ and $\varphi^\star \le \bar{\varphi}$. The second constraint implies $\beta/(2\sqrt{\lambda}) \le \bar{\varphi}$, or $\lambda \ge (\beta/2\bar{\varphi})^2$.

(2) If $\gamma \neq 1/2$, the interior condition cannot be satisfied. Therefore, no interior optimum exists, and the solution must lie on one of the boundaries as characterized in §3.2. □

B.3. Proofs for Constrained Design

This section provides the formal proofs for the propositions in Section 3.4 regarding operational and policy constraints.

Proof of Proposition 3.7 (When the Reliability Cap Binds). The provider solves max $\Phi(\varphi^\star, \lambda)$ sub-

ject to $\lambda \leq \bar{\lambda}$. The Lagrangian is $\mathcal{L} = \Phi(\varphi^\star, \lambda) + \eta(\bar{\lambda} - \lambda)$, with KKT conditions:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \lambda} &= F(\varphi^\star)(b'(\lambda) - \varphi^\star) - \eta = 0 \\ \frac{\partial \mathcal{L}}{\partial \varphi^\star} &= f(\varphi^\star)(b(\lambda) - \varphi^\star \lambda) - F(\varphi^\star)\lambda = 0 \\ \eta \geq 0, \quad \lambda \leq \bar{\lambda}, \quad \eta(\bar{\lambda} - \lambda) &= 0. \end{aligned}$$

If the cap binds, $\lambda = \bar{\lambda}$ and $\eta \geq 0$. The second KKT condition determines the optimal cutoff at this fixed hazard, $\varphi^\dagger(\bar{\lambda})$, which must solve $\frac{b(\bar{\lambda})}{\bar{\lambda}} - \varphi^\star = A(\varphi^\star)$. The first KKT condition implies $\eta = F(\varphi^\dagger(\bar{\lambda}))(b'(\bar{\lambda}) - \varphi^\dagger(\bar{\lambda}))$. For this to be a valid solution, we need $\eta \geq 0$, which requires $b'(\bar{\lambda}) \geq \varphi^\dagger(\bar{\lambda})$. This proves part (1). If $b'(\bar{\lambda}) < \varphi^\dagger(\bar{\lambda})$, then η would be negative, violating the KKT conditions. This means the constraint cannot be binding at the optimum. Since the unconstrained problem has a unique maximum (under regularity), the solution must be the unconstrained one, which must have $\lambda^\star < \bar{\lambda}$. This proves part (2). □

Proof of Proposition 3.8 (Effects of Binding Display Constraints). The problem is $\max \Phi(\varphi^\star, \lambda)$ subject to $\lambda(\varphi^\star - \underline{r}) \leq \chi$. The Lagrangian is $\mathcal{L} = \Phi(\varphi^\star, \lambda) + \tau(\chi - \lambda(\varphi^\star - \underline{r}))$. If the constraint binds, $\tau > 0$. The KKT conditions are:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \lambda} &= F(\varphi^\star)(b'(\lambda) - \varphi^\star) - \tau(\varphi^\star - \underline{r}) = 0 \\ \frac{\partial \mathcal{L}}{\partial \varphi^\star} &= f(\varphi^\star)(b(\lambda) - \varphi^\star \lambda) - F(\varphi^\star)\lambda - \tau\lambda = 0 \end{aligned}$$

From the first condition, since $\tau > 0$ and we assume $\varphi^\star > \underline{r}$, we must have $F(\varphi^\star)(b'(\lambda) - \varphi^\star) > 0$, which implies $b'(\lambda) > \varphi^\star$. At the unconstrained optimum, $b'(\lambda^M) = \varphi^M$. Since $b'(\cdot)$ is decreasing, for the inequality to hold, the constrained hazard must be lower than the unconstrained one. From the second condition, rearranging gives $\frac{b(\lambda)}{\lambda} - \varphi^\star = A(\varphi^\star) + \frac{\tau}{f(\varphi^\star)}$. The right-hand side is strictly greater than at the unconstrained optimum. Let $G(\varphi, \lambda) = \frac{b(\lambda)}{\lambda} - \varphi - A(\varphi)$. The unconstrained solution satisfies $G(\varphi^M, \lambda^M) = 0$. The constrained solution satisfies $G(\varphi^c, \lambda^c) = \frac{\tau}{f(\varphi^c)} > 0$. Since $\partial G/\partial \varphi = -1 - A'(\varphi) < 0$ and $\partial G/\partial \lambda < 0$, and we already know $\lambda^c < \lambda^M$, it must be that $\varphi^c < \varphi^M$ for the equality to hold. Both hazard and cutoff are lower. □

Appendix C: Proofs and Additional Material for Section 4

This appendix provides the formal proofs for the equivalence results presented in Section 4. We demonstrate rigorously that the spot instance menu is isomorphic to a canonical damaged-goods problem and that it implements the same allocation and expected payments as a simple priority auction.

C.1. Proof of Damaged-Goods Equivalence (Proposition 4.1)

The core of the argument is a re-parameterization of the choice variables to map our model into the standard vertical quality differentiation framework.

Proof of Proposition 4.1. We establish the equivalence by defining quality as reliability and consolidating pricing terms to match the standard damaged-goods formulation. Define:

- Quality: $q \equiv 1 - \lambda$ (so $q = 1$ for guaranteed, $q < 1$ for preemptible)
- Effective price: $\tilde{p}_P \equiv p_P - r\lambda$ (posted price minus expected refund)

The utility of a type-φ buyer from each contract becomes:

$$\begin{aligned} U_G(\varphi) &= v - p_G \\ U_P(\varphi) &= v - p_P - \varphi\lambda + r\lambda \\ &= v - (p_P - r\lambda) - \varphi\lambda \\ &= v - \tilde{p}_P - \varphi(1 - q) \end{aligned}$$

This is now in the standard form for vertical differentiation: utility equals gross value minus price minus type-dependent quality degradation cost. A buyer is indifferent between contracts when $U_G(\varphi^\star) = U_P(\varphi^\star)$:

$$\begin{aligned} v - p_G &= v - \tilde{p}_P - \varphi^\star(1 - q) \\ p_G - \tilde{p}_P &= \varphi^\star(1 - q) \\ \varphi^\star &= \frac{p_G - \tilde{p}_P}{1 - q} \end{aligned}$$

This shows the cutoff depends only on the effective price differential and quality gap, not on the specific decomposition of $\tilde{p}_P$ into p_P and r. By single-crossing (utility difference is monotonic in φ):

- Types $\varphi \geq \varphi^\star$ choose guaranteed: $D_G = 1 - F(\varphi^\star)$
- Types $\varphi < \varphi^\star$ choose preemptible: $D_P = F(\varphi^\star)$

The seller's profit per buyer is:

- Guaranteed: $\pi_G = p_G$ (normalized cost = 0)
- Preemptible: $\pi_P = p_P + b(\lambda) - r\lambda = \tilde{p}_P + b(1 - q)$

The term $b(1-q)$ represents the cost advantage from degradation (salvage value from interruptions). Total profit is:

$$\Pi = (1 - F(\varphi^\star))p_G + F(\varphi^\star)\big[\tilde{p}_P + b(1-q)\big]$$

As established in Proposition 2.1, optimal pricing sets $p_G^\star = v$ (IR binds for guaranteed buyers). From the indifference condition with $p_G = v$:

$$\begin{aligned} v - \tilde{p}_P &= \varphi^\star(1-q) \\ \tilde{p}_P &= v - \varphi^\star(1-q) \end{aligned}$$

Substituting both optimal prices into the profit function:

$$\begin{aligned} \Pi &= (1 - F(\varphi^\star))v + F(\varphi^\star)\big[(v - \varphi^\star(1-q)) + b(1-q)\big] \\ &= v - vF(\varphi^\star) + F(\varphi^\star)v - F(\varphi^\star)\varphi^\star(1-q) + F(\varphi^\star)b(1-q) \\ &= v + F(\varphi^\star)\big[b(1-q) - \varphi^\star(1-q)\big] \end{aligned}$$

Therefore, the seller's problem reduces to:

$$\max_{q\in[0,1],\ \varphi^\star\in[\underline{\varphi},\bar{\varphi}]} \Psi(\varphi^\star, q) \equiv F(\varphi^\star)\big[b(1-q) - \varphi^\star(1-q)\big]$$

This is identical to the original reduced problem $\Phi(\varphi^\star, \lambda) = F(\varphi^\star)[b(\lambda) - \varphi^\star\lambda]$ under the substitution $q = 1 - \lambda$. For an interior solution, we differentiate $\Psi(\varphi^\star, q)$. With respect to q:

$$\begin{aligned} \frac{\partial \Psi}{\partial q} &= F(\varphi^\star)\frac{\partial}{\partial q}\big[b(1-q) - \varphi^\star(1-q)\big] \\ &= F(\varphi^\star)\big[-b'(1-q)(-1) - \varphi^\star(-1)\big] \\ &= F(\varphi^\star)\big[b'(1-q) + \varphi^\star\big] \\ &= F(\varphi^\star)\big[\varphi^\star - b'(1-q)\big] \end{aligned}$$

Setting equal to zero (and using $F(\varphi^\star) > 0$ at interior):

$$\boxed{\varphi^\star = b'(1-q)}$$

This states that the marginal cost saving from further degradation equals the marginal buyer's

quality sensitivity. With respect to $\varphi^\star$:

$$\begin{aligned}\frac{\partial \Psi}{\partial \varphi^\star} &= f(\varphi^\star)\big[b(1-q) - \varphi^\star(1-q)\big] + F(\varphi^\star)\big[-(1-q)\big] \\ &= f(\varphi^\star)\big[b(1-q) - \varphi^\star(1-q)\big] - F(\varphi^\star)(1-q)\end{aligned}$$

Setting equal to zero and rearranging:

$$\begin{aligned}f(\varphi^\star)\big[b(1-q) - \varphi^\star(1-q)\big] &= F(\varphi^\star)(1-q) \\ \frac{b(1-q) - \varphi^\star(1-q)}{1-q} &= \frac{F(\varphi^\star)}{f(\varphi^\star)} \\ \boxed{\frac{b(1-q)}{1-q} - \varphi^\star = \frac{F(\varphi^\star)}{f(\varphi^\star)}}\end{aligned}$$

This is the standard monopoly screening condition: average profit margin equals the inverse hazard rate. **Converse (Implementability):** Given any target allocation $(\tilde{p}_P, q)$ that implements cutoff $\varphi^\star$, we can achieve it through multiple (p_P, r) pairs.

From $\tilde{p}_P = p_P - r\lambda$ and $\lambda = 1 - q$:

$$p_P = \tilde{p}_P + r(1-q)$$

For any refund $r \geq 0$, we can set $p_P = \tilde{p}_P + r(1-q)$. All such pairs:

- Implement the same cutoff (since only $\tilde{p}_P$ matters for buyer decisions)
- Generate the same profit (since $\pi_P = \tilde{p}_P + b(1-q)$ regardless of r)
- Give buyers the same utilities

This confirms that refunds are purely implementational instruments. □

Proof of Corollary 4.1. If $b(\lambda) \equiv 0$ for all λ, then $b(1-q) \equiv 0$ for all q. The seller's objective in quality space becomes:

$$\begin{aligned}\max_{q,\varphi^\star} \Psi(\varphi^\star, q) &= F(\varphi^\star)\big[b(1-q) - \varphi^\star(1-q)\big] \\ &= F(\varphi^\star)\big[0 - \varphi^\star(1-q)\big] \\ &= -F(\varphi^\star)\varphi^\star(1-q)\end{aligned}$$

Since $F(\varphi^\star) \geq 0$, $\varphi^\star \geq \underline{\varphi} \geq 0$, and $(1-q) \geq 0$, the objective is non-positive. To maximize (make least negative), the seller must set either:

- $F(\varphi^\star) = 0$ (but this means no preemptible sales)
- $\varphi^\star = 0$ (but typically $\underline{\varphi} > 0$)
- $(1 - q) = 0$, which means $q = 1$ (no degradation)

The only feasible solution is $q^\star = 1$ (equivalently, $\lambda^\star = 0$), yielding $\Psi = 0$. The menu collapses to a single guaranteed product at price $p_G = v$, with profit $\Pi = v$. Without cost advantages from degradation, offering a damaged version only cannibalizes revenue from the premium product. The firm optimally offers only the high-quality version. □

Proof of Corollary 4.2. We show that introducing a small amount of degradation ($\lambda > 0$ or $q < 1$) strictly improves profit when $b'(0^+) > \underline{\varphi}$. Consider the profit gain from degradation:

$$\Psi(\varphi^\star, 1 - \lambda) = F(\varphi^\star)\big[b(\lambda) - \varphi^\star\lambda\big]$$

For small $\lambda > 0$, using Taylor expansion around $\lambda = 0$:

$$b(\lambda) = b(0) + b'(0^+)\lambda + o(\lambda) = b'(0^+)\lambda + o(\lambda)$$

since $b(0) = 0$ by assumption. Therefore:

$$\begin{aligned}\Psi(\varphi^\star, 1 - \lambda) &= F(\varphi^\star)\big[b'(0^+)\lambda + o(\lambda) - \varphi^\star\lambda\big] \\ &= F(\varphi^\star)\lambda\big[b'(0^+) - \varphi^\star + \frac{o(\lambda)}{\lambda}\big]\end{aligned}$$

Given $b'(0^+) > \underline{\varphi}$, we can choose $\varphi^\star \in (\underline{\varphi}, b'(0^+))$. For this choice:

- $F(\varphi^\star) > 0$ (positive mass of buyers below the cutoff)
- $b'(0^+) - \varphi^\star > 0$ (positive margin)
- As $\lambda \to 0^+$, the term $o(\lambda)/\lambda \to 0$

Therefore, for sufficiently small $\lambda > 0$:

$$\Psi(\varphi^\star, 1 - \lambda) > 0$$

This exceeds the profit from no degradation ($\Psi = 0$ when $\lambda = 0$). **Economic interpretation:** When the marginal value of initial capacity reclamation exceeds the lowest customer's interruption cost, the firm can profitably segment the market by offering a slightly degraded version. The operational benefit (salvage value) exceeds the revenue loss from degradation. □

The equivalence established here satisfies all conditions for the Revenue Equivalence Theorem:

1. **Monotonic allocation:** Higher types get higher quality (guaranteed service)
2. **Same allocation rule:** Both mechanisms assign types $\varphi \geq \varphi^\star$ to guaranteed service
3. **Same utility for lowest type:** $U(\underline{\varphi}) = \lambda(\varphi^\star - \underline{\varphi})$ in both
4. **Envelope condition:** $U'(\varphi) = -\lambda$ for $\varphi < \varphi^\star$ in both mechanisms

Therefore, the revenue equivalence between the posted menu and the damaged-goods formulation is rigorously established.

C.2. Proof of Priority Auction Equivalence (Proposition 4.2)

The equivalence between the posted menu and a priority auction is established through the revenue equivalence principle. We first derive the utility schedule for any truthful priority auction with a cutoff rule and then show that our posted menu implements the exact same schedule.

Lemma .1 (Envelope Theorem for Priority Auctions)**.** *Consider any truthful (Direct, Incentive-Compatible, Individually Rational) mechanism that assigns guaranteed service to types $\varphi \geq \varphi^\star$ and preemptible service (with hazard schedule h_κ) to types $\varphi < \varphi^\star$. Let $U(\varphi)$ be the equilibrium expected utility of type φ. With the normalization $U(\varphi^\star) = 0$, the utility schedule must be:*

$$U(\varphi) = \lambda(\kappa)(\varphi^\star - \varphi) \quad \textit{for } \varphi < \varphi^\star, \quad \textit{and} \quad U(\varphi) = 0 \quad \textit{for } \varphi \geq \varphi^\star.$$

Proof. Let $x(\varphi) = 1$ if type φ gets guaranteed service and $x(\varphi) = 0$ if they get preemptible service. By assumption, $x(\varphi) = \mathbf{1}\{\varphi \geq \varphi^\star\}$. The utility is $v - p - \varphi\lambda$ for P and $v - p_G$ for G. The type-dependent part of the utility is $-\varphi\lambda$. The envelope theorem states $U'(\varphi) = \frac{\partial U}{\partial \varphi}|_{\hat{\varphi}=\varphi}$. For a buyer in the preemptible class, the utility is $U_P(\varphi)$. Her reported type $\hat{\varphi}$ determines her allocation and payment, but her true utility depends on her true type φ. The utility function is of the form $k(\hat{\varphi}) - \varphi \cdot q(\hat{\varphi})$. Here, $q(\hat{\varphi}) = \lambda$ if $\hat{\varphi} < \varphi^\star$ and 0 otherwise. So $U'(\varphi) = -q(\varphi)$. Thus, $U'(\varphi) = -\lambda(\kappa)$ for $\varphi < \varphi^\star$ and $U'(\varphi) = 0$ for $\varphi \geq \varphi^\star$. Integrating this from $\varphi^\star$ downwards gives:

$$U(\varphi) - U(\varphi^\star) = \int_{\varphi^\star}^{\varphi} (-\lambda(\kappa))dt = \lambda(\kappa)(\varphi^\star - \varphi) \quad \text{for } \varphi < \varphi^\star.$$

With the normalization $U(\varphi^\star) = 0$ (which follows from setting $p_G = v$ and ensuring the marginal buyer participates), we get the desired result. For $\varphi \geq \varphi^\star$, $U'(\varphi) = 0$, so $U(\varphi)$ is constant and equal to $U(\varphi^\star) = 0$. □

Proof of Proposition 4.2. The proof is now a direct comparison. **Auction $\implies$ Menu**: Given a truthful auction with cutoff $\varphi^\star$ and hazard $\lambda(\kappa)$, Lemma .1 shows it must generate the utility

schedule $U(\varphi) = \lambda(\kappa)(\varphi^\star - \varphi)_+$. Now consider our posted menu. Set $p_G = v$. This immediately implies $U_G(\varphi) = 0$ for all $\varphi \geq \varphi^\star$. For the preemptible contract, choose any refund r and set the price $p_P = v - (\varphi^\star - r)\lambda(\kappa)$. The utility for a type $\varphi < \varphi^\star$ is:

$$U_P(\varphi) = v - p_P - \varphi\lambda(\kappa) + r\lambda(\kappa) = v - \Big(v - (\varphi^\star - r)\lambda(\kappa)\Big) - \varphi\lambda(\kappa) + r\lambda(\kappa) = \lambda(\kappa)(\varphi^\star - \varphi).$$

The posted menu perfectly replicates the utility schedule of the auction. Since both mechanisms result in the same allocation (same cutoff $\varphi^\star$) and give every agent the same expected utility, the Revenue Equivalence Theorem implies they must generate the same expected profit for the seller.

Menu $\implies$ Auction: The logic is symmetric. Given an optimal posted menu $(\lambda^\star, \varphi^\star)$ implemented with some (p_G, p_P, r), we know it generates a certain utility schedule. We can construct a direct revelation mechanism with the same allocation rule (cutoff at $\varphi^\star$) and the same hazard schedule ($\lambda^\star$). We can then define a transfer function $T(\varphi)$ that ensures truth-telling and generates the exact same utility schedule as the posted menu. By revenue equivalence, this auction will be profit-equivalent. □

Proof of Corollary 4.3. The optimal interior menu is characterized by the two first-order conditions in Proposition 3.1. The first is $\varphi^\star = b'(\lambda)$. From the salvage microfoundation (Proposition .2), we know that $b'(\lambda) = \kappa(\lambda)$. Therefore, at the optimum, we must have $\varphi^\star = \kappa(\lambda^\star)$. This means the market-based cutoff for priority must be aligned with the operational cutoff for interruptions. Proposition 4.2 then guarantees that this allocation, which is optimal for the monopolist under a posted menu, is also implementable via a priority auction with these aligned thresholds. □

Appendix D: Proofs and Additional Material for Section 5

This appendix provides the formal proofs for the welfare analysis and policy evaluations presented in Section 5. We maintain the primitives and notation from the main text. The monopolist's reduced profit function is $\Phi(\varphi^\star, \lambda) = F(\varphi^\star)\big(b(\lambda) - \varphi^\star\lambda\big)$, and the planner's baseline welfare function is $W(\varphi^\star, \lambda) = v + F(\varphi^\star)\big(b(\lambda) - m(\varphi^\star)\lambda\big)$, where $m(\varphi^\star) \equiv \mathbb{E}[\varphi \mid \varphi < \varphi^\star]$.

D.1. The Planner's Problem: Proof of Proposition 5.1

This section provides a detailed derivation of the first-order conditions that characterize the utilitarian social planner's optimal allocation.

Lemma .2 (Derivatives of the Welfare Function)**.** *For $\lambda \in (0, 1]$ and $\varphi^\star \in (\underline{\varphi}, \bar{\varphi})$, the partial derivatives of the welfare function $W(\varphi^\star, \lambda)$ are:*

$$\frac{\partial W}{\partial \lambda} = F(\varphi^\star)\big(b'(\lambda) - m(\varphi^\star)\big),$$

$$\frac{\partial W}{\partial \varphi^\star} = f(\varphi^\star)\big(b(\lambda) - \varphi^\star \lambda\big).$$

Proof. The welfare function is $W(\varphi^\star, \lambda) = v + F(\varphi^\star)b(\lambda) - \lambda F(\varphi^\star)m(\varphi^\star)$. The derivative with respect to λ is straightforward:

$$\frac{\partial W}{\partial \lambda} = F(\varphi^\star)b'(\lambda) - F(\varphi^\star)m(\varphi^\star) = F(\varphi^\star)\big(b'(\lambda) - m(\varphi^\star)\big).$$

For the derivative with respect to $\varphi^\star$, we first note the identity that follows from the definition of the conditional mean $m(\varphi^\star) = \frac{1}{F(\varphi^\star)} \int_{\underline{\varphi}}^{\varphi^\star} t f(t) dt$. Differentiating the term $F(\varphi^\star)m(\varphi^\star) = \int_{\underline{\varphi}}^{\varphi^\star} t f(t) dt$ with respect to $\varphi^\star$ using Leibniz's rule gives:

$$\frac{d}{d\varphi^\star}\Big(F(\varphi^\star)m(\varphi^\star)\Big) = \varphi^\star f(\varphi^\star).$$

Now, we differentiate the welfare function with respect to $\varphi^\star$:

$$\frac{\partial W}{\partial \varphi^\star} = f(\varphi^\star)b(\lambda) - \lambda \frac{d}{d\varphi^\star}\Big(F(\varphi^\star)m(\varphi^\star)\Big) = f(\varphi^\star)b(\lambda) - \lambda\varphi^\star f(\varphi^\star) = f(\varphi^\star)\big(b(\lambda) - \varphi^\star\lambda\big).$$

This completes the derivation of the partial derivatives. □

Proof of Proposition 5.1. An interior social optimum $(\varphi^{\star\,\mathrm{fb}}, \lambda^{\mathrm{fb}})$ must satisfy the first-order conditions $\frac{\partial W}{\partial \lambda} = 0$ and $\frac{\partial W}{\partial \varphi^\star} = 0$. Using the derivatives from Lemma .2:

$$F(\varphi^{\star\,\mathrm{fb}})\big(b'(\lambda^{\mathrm{fb}}) - m(\varphi^{\star\,\mathrm{fb}})\big) = 0 \implies b'(\lambda^{\mathrm{fb}}) = m(\varphi^{\star\,\mathrm{fb}}),$$

since $F(\varphi^{\star\,\mathrm{fb}}) > 0$ at an interior solution. And:

$$f(\varphi^{\star\,\mathrm{fb}})\big(b(\lambda^{\mathrm{fb}}) - \varphi^{\star\,\mathrm{fb}}\lambda^{\mathrm{fb}}\big) = 0 \implies b(\lambda^{\mathrm{fb}}) = \varphi^{\star\,\mathrm{fb}}\lambda^{\mathrm{fb}} \implies \frac{b(\lambda^{\mathrm{fb}})}{\lambda^{\mathrm{fb}}} = \varphi^{\star\,\mathrm{fb}},$$

since $f(\varphi^{\star\,\mathrm{fb}}) > 0$. These are the two conditions stated in the proposition. The concavity of W in λ (since $b'' \leq 0$) and the single-peakedness of W in $\varphi^\star$ (since $b(\lambda)/\lambda - \varphi^\star$ is strictly decreasing in $\varphi^\star$) ensure that these conditions are sufficient for a local maximum under standard regularity. □

Proof of Monopoly Distortions. The proof is by direct comparison of the first-order conditions. **Enrollment Distortion**: The planner's cutoff rule is $\varphi^{fb} = b(\lambda^{fb})/\lambda^{fb}$. The monopolist's is $\varphi^M = b(\lambda^M)/\lambda^M - A(\varphi^M)$, where $A(\varphi^M) = F(\varphi^M)/f(\varphi^M) > 0$. This implies $\varphi^M < b(\lambda^M)/\lambda^M$. Let's define the functions $\varphi_P(\lambda) = b(\lambda)/\lambda$ (the planner's optimal cutoff for a given λ) and $\varphi_M(\lambda)$ as the solution to $\varphi_M = b(\lambda)/\lambda - A(\varphi_M)$. Since $A(\cdot) > 0$, for any λ, we have $\varphi_M(\lambda) < \varphi_P(\lambda)$. As both $\varphi_P(\lambda)$ and $\varphi_M(\lambda)$ are typically decreasing in λ (due to concavity of b), comparing the full solutions

requires solving the systems. However, the structure of the equations points toward $\varphi^M < \varphi^{fb}$. At the monopoly solution (φ^M, λ^M), the welfare derivative $\partial W/\partial\varphi^\star = f(\varphi^M)(b(\lambda^M) - \varphi^M\lambda^M) = f(\varphi^M)\lambda^M A(\varphi^M) > 0$, indicating that welfare would be increased by raising the cutoff.

Hazard Distortion: The monopolist's hazard rule is $b'(\lambda^M) = \varphi^M$. The planner's is $b'(\lambda^{fb}) = m(\varphi^{fb})$. For any given cutoff φ, the monopolist would set λ such that $b'(\lambda) = \varphi$, while the planner would set it such that $b'(\lambda) = m(\varphi)$. Since $\varphi > m(\varphi)$ and $b'(\cdot)$ is a decreasing function, the hazard chosen by the planner for that cutoff would be higher than the one chosen by the monopolist. At the monopoly solution, the welfare derivative $\partial W/\partial\lambda = F(\varphi^M)(b'(\lambda^M) - m(\varphi^M)) = F(\varphi^M)(\varphi^M - m(\varphi^M)) > 0$, indicating that welfare would be increased by raising the hazard, holding the cutoff fixed. □

D.2. Policy Instruments: Proofs and Derivations

This section contains the formal proofs for the policy analysis in Section 5.2. We provide complete derivations for the sufficient-statistic transfer and the welfare effects of various policy instruments.

Proof of Proposition 5.2 (Sufficient-Statistic Transfer). We construct a transfer function that induces the monopolist to choose the planner's allocation. The approach is to design a transfer $\mathcal{T}(\lambda, D)$ depending only on observables such that the monopolist's first-order conditions coincide with the planner's.

The monopolist maximizes regulated profit:

$$\Pi^{\text{reg}}(\varphi^\star, \lambda) = F(\varphi^\star)\big[b(\lambda) - \varphi^\star\lambda\big] + \mathcal{T}\big(\lambda, F(\varphi^\star)\big)$$

The first-order conditions for the regulated monopolist are:

$$\frac{\partial\Pi^{\text{reg}}}{\partial\lambda} = F(\varphi^\star)\big[b'(\lambda) - \varphi^\star\big] + \frac{\partial\mathcal{T}}{\partial\lambda} = 0 \tag{28}$$

$$\frac{\partial\Pi^{\text{reg}}}{\partial\varphi^\star} = f(\varphi^\star)\big[b(\lambda) - \varphi^\star\lambda\big] - F(\varphi^\star)\lambda + f(\varphi^\star)\frac{\partial\mathcal{T}}{\partial D} = 0 \tag{29}$$

The planner's first-order conditions (from Proposition 5.1) are:

$$b'(\lambda^{\text{fb}}) = m(\varphi^{\star\,\text{fb}}) \tag{30}$$

$$\frac{b(\lambda^{\text{fb}})}{\lambda^{\text{fb}}} = \varphi^{\star\,\text{fb}} \tag{31}$$

For the regulated monopolist's FOCs to match the planner's, we need to determine what $\frac{\partial\mathcal{T}}{\partial\lambda}$ and $\frac{\partial\mathcal{T}}{\partial D}$ must be.

From equation (28), if we want $b'(\lambda) = m(\varphi^\star)$ to hold:

$$\begin{aligned} F(\varphi^\star)\big[b'(\lambda) - \varphi^\star\big] + \frac{\partial \mathcal{T}}{\partial \lambda} &= 0 \\ F(\varphi^\star)\big[m(\varphi^\star) - \varphi^\star\big] + \frac{\partial \mathcal{T}}{\partial \lambda} &= 0 \\ \therefore \quad \frac{\partial \mathcal{T}}{\partial \lambda} &= F(\varphi^\star)\big[\varphi^\star - m(\varphi^\star)\big] \end{aligned}$$

From equation (29), dividing by $f(\varphi^\star) > 0$:

$$\big[b(\lambda) - \varphi^\star \lambda\big] - \frac{F(\varphi^\star)}{f(\varphi^\star)}\lambda + \frac{\partial \mathcal{T}}{\partial D} = 0$$

If we want $b(\lambda) = \varphi^\star \lambda$ to hold (the planner's cutoff rule):

$$\begin{aligned} 0 - A(\varphi^\star)\lambda + \frac{\partial \mathcal{T}}{\partial D} &= 0 \\ \therefore \quad \frac{\partial \mathcal{T}}{\partial D} &= A(\varphi^\star)\lambda \end{aligned}$$

where $A(\varphi) \equiv F(\varphi)/f(\varphi)$ is the inverse hazard rate.

For a function $\mathcal{T}(\lambda, D)$ to exist with the required partial derivatives, we need the cross-partials to be equal:

$$\frac{\partial^2 \mathcal{T}}{\partial D \partial \lambda} = \frac{\partial^2 \mathcal{T}}{\partial \lambda \partial D}$$

Let $\varphi(D) = F^{-1}(D)$ denote the inverse CDF. Then $D = F(\varphi)$ and $\frac{d\varphi}{dD} = \frac{1}{f(\varphi)}$.

Computing the left side:

$$\begin{aligned} \frac{\partial}{\partial D}\Big[D\big(\varphi(D) - m(\varphi(D))\big)\Big] &= \big(\varphi - m(\varphi)\big) + D\frac{d}{dD}\big[\varphi - m(\varphi)\big] \\ &= \big(\varphi - m\big) + D \cdot \frac{1}{f(\varphi)} \cdot \frac{d}{d\varphi}\big[\varphi - m(\varphi)\big] \end{aligned}$$

We need to compute $\frac{dm}{d\varphi}$. By definition, $m(\varphi) = \frac{1}{F(\varphi)} \int_{\underline{\varphi}}^{\varphi} t f(t) dt$. Using the quotient rule and Leibniz's rule:

$$\frac{dm}{d\varphi} = \frac{\varphi f(\varphi) \cdot F(\varphi) - f(\varphi) \int_{\underline{\varphi}}^{\varphi} t f(t) dt}{F(\varphi)^2} = \frac{f(\varphi)}{F(\varphi)}\big[\varphi - m(\varphi)\big]$$

Continuing the computation:

$$\begin{aligned}\frac{\partial}{\partial D}\Big[D\big(\varphi - m\big)\Big] &= \big(\varphi - m\big) + \frac{D}{f(\varphi)} \cdot \left[1 - \frac{f(\varphi)}{F(\varphi)}\big(\varphi - m\big)\right] \\ &= \big(\varphi - m\big) + \frac{D}{f(\varphi)} - \frac{D}{F(\varphi)}\big(\varphi - m\big) \\ &= \big(\varphi - m\big)\left[1 - \frac{D}{F(\varphi)}\right] + \frac{D}{f(\varphi)} \\ &= 0 + \frac{F(\varphi)}{f(\varphi)} = A(\varphi)\end{aligned}$$

Computing the right side:

$$\frac{\partial}{\partial \lambda}\Big[A(\varphi(D))\lambda\Big] = A(\varphi(D))$$

Since both cross-partials equal $A(\varphi(D))$, the integrability condition is satisfied.

We can integrate either partial derivative to construct $\mathcal{T}$. Integrating $\frac{\partial \mathcal{T}}{\partial \lambda} = D[\varphi(D) - m(\varphi(D))]$ with respect to λ:

$$\mathcal{T}(\lambda, D) = \lambda D\Big[\varphi(D) - m(\varphi(D))\Big] + g(D)$$

where $g(D)$ is an integration "constant" that may depend on D. To find $g(D)$, we use the condition $\frac{\partial \mathcal{T}}{\partial D} = A(\varphi(D))\lambda$:

$$\begin{aligned}\frac{\partial \mathcal{T}}{\partial D} &= \lambda \frac{\partial}{\partial D}\Big[D\big(\varphi(D) - m(\varphi(D))\big)\Big] + g'(D) \\ &= \lambda A(\varphi(D)) + g'(D)\end{aligned}$$

This must equal $A(\varphi(D))\lambda$, so $g'(D) = 0$, implying $g(D) = C$ for some constant C. Therefore, a valid transfer function is:

$$\boxed{\mathcal{T}(\lambda, D) = \lambda D\Big[\varphi(D) - m(\varphi(D))\Big] + C}$$

where $\varphi(D) = F^{-1}(D)$ and $m(\varphi) = \mathbb{E}[\tilde{\varphi} \mid \tilde{\varphi} < \varphi]$.

With this transfer in place:

1. The monopolist's FOCs become exactly the planner's FOCs, inducing the first-best allocation $(\varphi^{\text{fb}}, \lambda^{\text{fb}})$.

2. The transfer depends only on observables: the hazard λ, the market share $D = F(\varphi^\star)$, and the known distribution F.

3. The regulator can compute and commit to this transfer schedule ex-ante.

The transfer $\mathcal{T}$ provides a per-user subsidy of $\lambda[\varphi - m(\varphi)]$ to the preemptible segment. Since $\varphi > m(\varphi)$, this is positive and encourages the monopolist to increase the hazard (lower reliability) toward the socially optimal level. The subsidy is proportional to the gap between marginal and average disutility, precisely correcting the monopolist's incentive distortion.

The constant C can be chosen to satisfy any budget constraint or to ensure the transfer is revenue-neutral in expectation. □

Proof of Proposition 5.3 (Welfare Effect of Reliability Caps). We analyze how a marginal tightening of a reliability cap affects welfare, starting from the unconstrained monopoly allocation.

Let $W(\varphi^\star(\lambda), \lambda)$ be welfare along the monopolist's best-response function $\varphi^\star(\lambda)$. The total derivative with respect to λ is:

$$\frac{dW}{d\lambda} = \frac{\partial W}{\partial \lambda} + \frac{\partial W}{\partial \varphi^\star} \cdot \frac{d\varphi^\star}{d\lambda}$$

From equations (20) and (21):

$$\left.\frac{\partial W}{\partial \lambda}\right|_{\mathrm{M}} = F(\varphi^{\mathrm{M}})\left[\varphi^{\mathrm{M}} - m(\varphi^{\mathrm{M}})\right]$$
$$\left.\frac{\partial W}{\partial \varphi^\star}\right|_{\mathrm{M}} = f(\varphi^{\mathrm{M}})\lambda^{\mathrm{M}} A(\varphi^{\mathrm{M}})$$

The monopolist's cutoff rule is $H(\varphi^\star, \lambda) \equiv \frac{b(\lambda)}{\lambda} - \varphi^\star - A(\varphi^\star) = 0$. Implicitly differentiating:

$$\frac{\partial H}{\partial \lambda} + \frac{\partial H}{\partial \varphi^\star} \cdot \frac{d\varphi^\star}{d\lambda} = 0$$

Computing the partial derivatives:

$$\frac{\partial H}{\partial \lambda} = \frac{b'(\lambda)\lambda - b(\lambda)}{\lambda^2}$$
$$\frac{\partial H}{\partial \varphi^\star} = -1 - A'(\varphi^\star)$$

At the monopoly point where $b'(\lambda^{\mathrm{M}}) = \varphi^{\mathrm{M}}$ and $\frac{b(\lambda^{\mathrm{M}})}{\lambda^{\mathrm{M}}} = \varphi^{\mathrm{M}} + A(\varphi^{\mathrm{M}})$:

$$\left.\frac{\partial H}{\partial \lambda}\right|_{\mathrm{M}} = \frac{\varphi^{\mathrm{M}}\lambda^{\mathrm{M}} - (\varphi^{\mathrm{M}} + A(\varphi^{\mathrm{M}}))\lambda^{\mathrm{M}}}{(\lambda^{\mathrm{M}})^2} = \frac{-A(\varphi^{\mathrm{M}})}{\lambda^{\mathrm{M}}}$$

Therefore:

$$\left.\frac{d\varphi^\star}{d\lambda}\right|_{\mathrm{M}} = -\frac{\partial H/\partial \lambda}{\partial H/\partial \varphi^\star} = -\frac{A(\varphi^{\mathrm{M}})/\lambda^{\mathrm{M}}}{1 + A'(\varphi^{\mathrm{M}})} = -\frac{A(\varphi^{\mathrm{M}})}{\lambda^{\mathrm{M}}[1 + A'(\varphi^{\mathrm{M}})]}$$

Total welfare derivative is:

$$\begin{aligned}\left.\frac{dW}{d\lambda}\right|_{\mathrm{M}} &= F(\varphi^{\mathrm{M}})\left[\varphi^{\mathrm{M}} - m(\varphi^{\mathrm{M}})\right] - f(\varphi^{\mathrm{M}})\lambda^{\mathrm{M}}A(\varphi^{\mathrm{M}}) \cdot \frac{A(\varphi^{\mathrm{M}})}{\lambda^{\mathrm{M}}[1 + A'(\varphi^{\mathrm{M}})]} \\ &= F(\varphi^{\mathrm{M}})\left[\varphi^{\mathrm{M}} - m(\varphi^{\mathrm{M}}) - \frac{A(\varphi^{\mathrm{M}})^2}{1 + A'(\varphi^{\mathrm{M}})} \cdot \frac{f(\varphi^{\mathrm{M}})}{F(\varphi^{\mathrm{M}})}\right] \\ &= F(\varphi^{\mathrm{M}})\left[\varphi^{\mathrm{M}} - m(\varphi^{\mathrm{M}}) - \frac{A(\varphi^{\mathrm{M}})}{1 + A'(\varphi^{\mathrm{M}})}\right]\end{aligned}$$

A marginal tightening of the cap (reducing $\bar{\lambda}$) increases welfare if $\frac{dW}{d\lambda} < 0$, which occurs if and only if:

$$\boxed{\varphi^{\mathrm{M}} - m(\varphi^{\mathrm{M}}) < \frac{A(\varphi^{\mathrm{M}})}{1 + A'(\varphi^{\mathrm{M}})}}$$

This condition compares the "interruption wedge" (left side) with a modified "screening wedge" (right side). □

Proof of Proposition 5.4 (Welfare Effect of Display Rules). This proof is more straightforward than the reliability cap case.

At the unconstrained monopoly allocation $(\varphi^{\mathrm{M}}, \lambda^{\mathrm{M}})$, we established that:

$$\begin{aligned}\left.\frac{\partial W}{\partial \lambda}\right|_{\mathrm{M}} &= F(\varphi^{\mathrm{M}})\left[\varphi^{\mathrm{M}} - m(\varphi^{\mathrm{M}})\right] > 0 \\ \left.\frac{\partial W}{\partial \varphi^{\star}}\right|_{\mathrm{M}} &= f(\varphi^{\mathrm{M}})\lambda^{\mathrm{M}}A(\varphi^{\mathrm{M}}) > 0\end{aligned}$$

From Proposition 3.8, a binding display constraint forces the monopolist to choose both a lower hazard and a lower cutoff: $\lambda^{c} < \lambda^{\mathrm{M}}$ and $\varphi^{c} < \varphi^{\mathrm{M}}$.

The first-order change in welfare is:

$$dW = \left.\frac{\partial W}{\partial \lambda}\right|_{\mathrm{M}} \underbrace{(d\lambda)}_{<0} + \left.\frac{\partial W}{\partial \varphi^{\star}}\right|_{\mathrm{M}} \underbrace{(d\varphi^{\star})}_{<0} < 0$$

Since both partial derivatives are positive and both changes are negative, the total welfare change is unambiguously negative. Therefore, tightening a display constraint is locally welfare-reducing. □

D.3. Analysis with Externalities

The proofs here extend the planner's problem and the policy analysis to include a spillover cost term $\Xi(\lambda, D)$.

Derivation of Planner's FOCs with Externalities, Eq. (24). The planner's objective is $W^{\text{ext}} = W(\varphi^\star, \lambda) - \Xi(\lambda, F(\varphi^\star))$. We add the derivatives of the externality term to the derivatives from Lemma .2.

$$\frac{\partial W^{\text{ext}}}{\partial \lambda} = \frac{\partial W}{\partial \lambda} - \frac{\partial \Xi}{\partial \lambda} = F(\varphi^\star)(b'(\lambda) - m(\varphi^\star)) - \Xi_\lambda.$$

Setting this to zero gives $b'(\lambda) = m(\varphi^\star) + \Xi_\lambda / F(\varphi^\star)$.

$$\frac{\partial W^{\text{ext}}}{\partial \varphi^\star} = \frac{\partial W}{\partial \varphi^\star} - \frac{\partial \Xi}{\partial D} f(\varphi^\star) = f(\varphi^\star)(b(\lambda) - \varphi^\star \lambda) - \Xi_D f(\varphi^\star).$$

Setting this to zero gives $b(\lambda)/\lambda = \varphi^\star + \Xi_D/\lambda$. (Assuming Ξ is a cost, so it enters with a minus sign). □